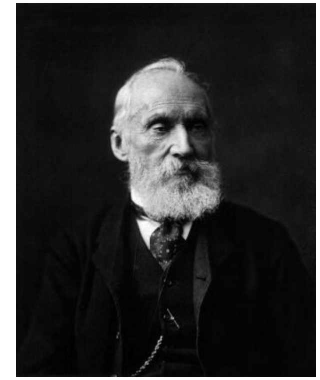
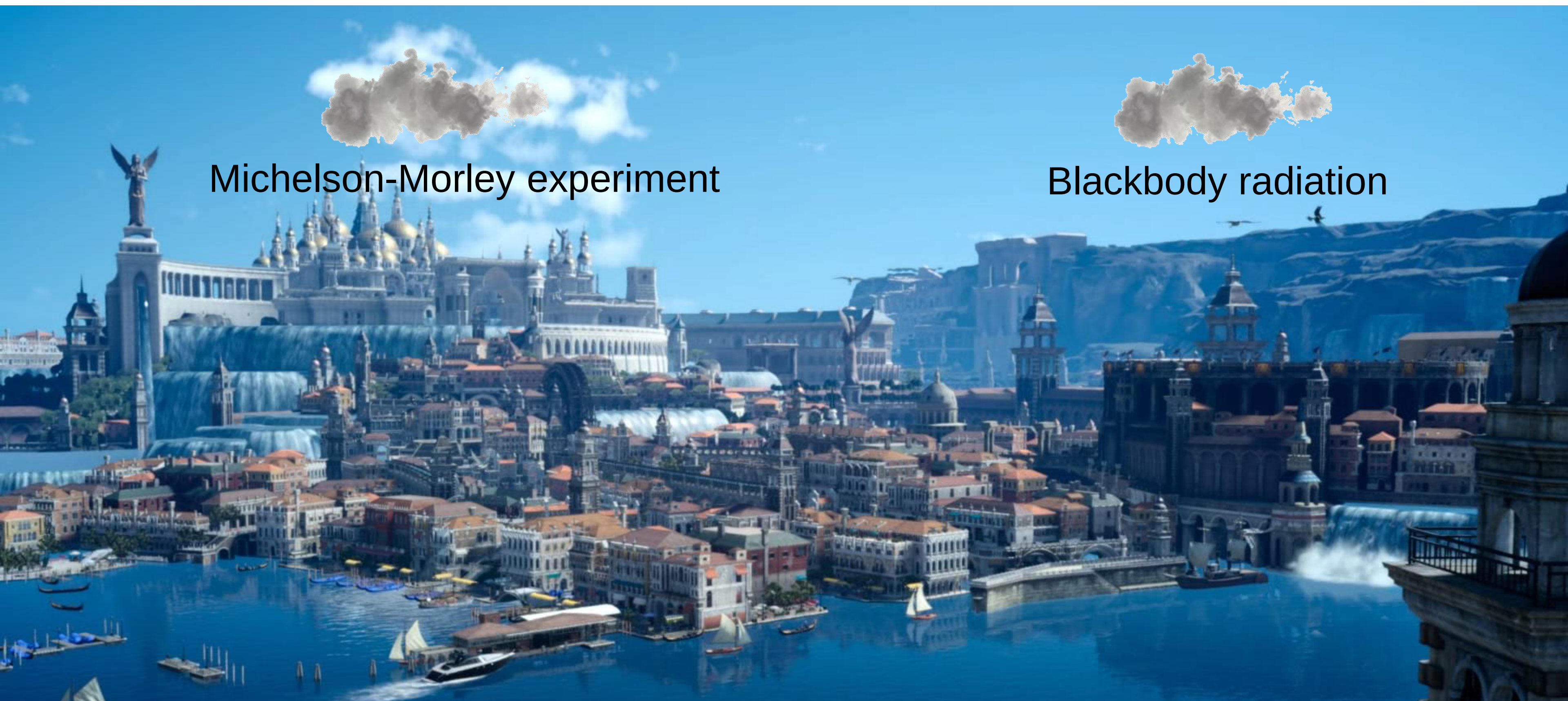
Two small, puzzling clouds

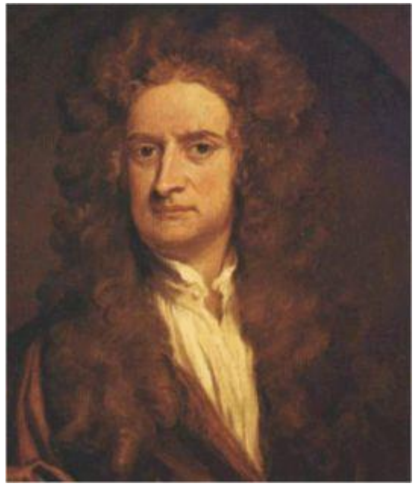
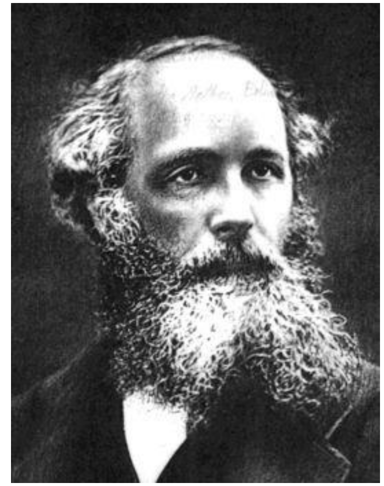
April 27, 1900, Royal Society
William Thomson (Baron Kelvin, 1824~1907)



There is nothing new to be discovered in physics now. All that remains is more and more precise measurement.
But…
<u>two small, puzzling clouds remained on the horizon.</u>



Physics at the turn of the 20th century

The beauty and clearness of the dynamical theory, which asserts heat and light to be modes of motion, is at present obscured by **two clouds**.

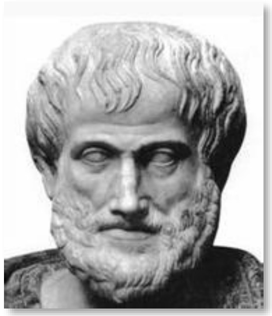
--- Lord Kelvin

Two clouds:

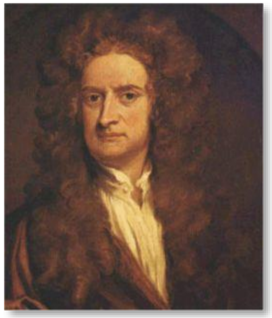
1. The inability to detect the luminous ether, specifically the failure of the Michelson-Morley experiment.

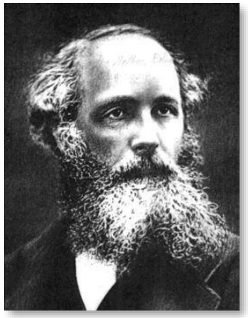
2. The black body radiation effect known as the ultraviolet catastrophe.

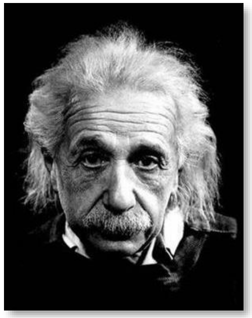
The myth of ether…


Aristotle


Descartes


Newton


Maxwell


Einstein

The concept of ether was first proposed by Aristotle as one of the five classical elements: earth, water, air, fire, and **ether** (or aether).

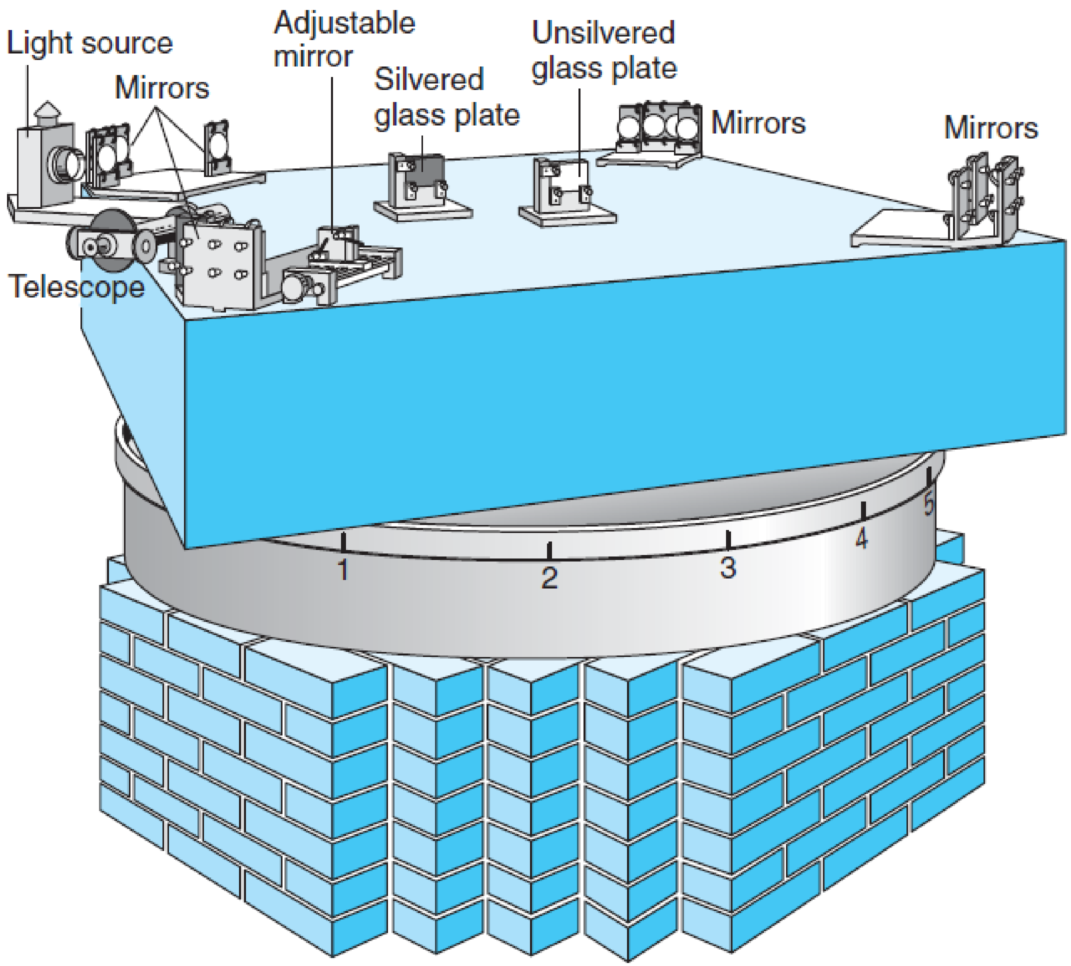
Descartes developed the idea of ether by attributing mechanical quality to it. According to Descartes, ether mediates interaction among objects and there is no action at distance.

Newton adopted Descartes' idea of ether as a mediator of interaction. Ether also forms the absolute space frame required by the Newtonian mechanics.

Maxwell also believed in the existence of ether, which he thought was the medium for electromagnetic waves.

Einstein's special theory of relativity finally **ruled out** the existence of ether.

Michelson-Morley's experiment in 1887



Speed of light is the same in all directions!

Modern Physics vs classical physics

In very brief terms :

Classical physics is physics based on principles describing **macroscopic** system and **low speed** physical systems.

Modern physics, in contrast to classical physics, deals with **microscopic** and **high speed** (comparable to the speed of light) situations. Modern physics may refer to: **Quantum mechanics** and **Theory of relativity**.

Macroscopic physical system means the scale of atoms and molecules on upwards, because inside the atom and among atoms in a molecule, the laws of classical physics break down and generally do not provide a correct description. **Quantum physics** will be the correct theory to approach.

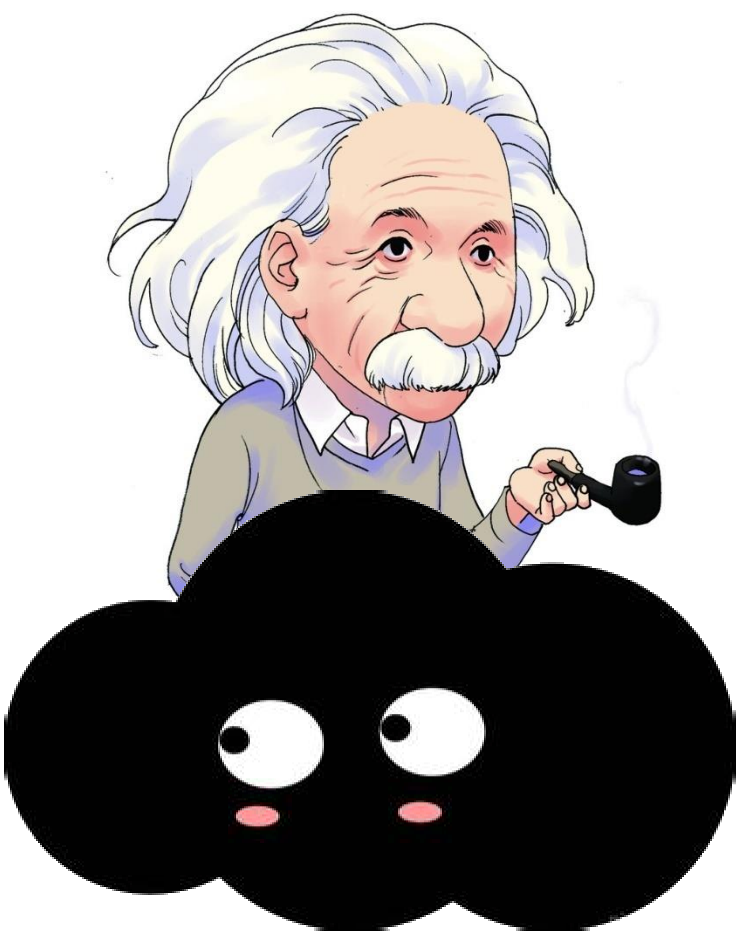
Low speed system means the speed is far below the speed of light $v \ll c$. The laws of classical physics break down when speed is closer to the speed of light, $v \le c$, and Einstein's **theory of relativity** with be the right theory to approach.

Modern Physics vs classical physics

Classical physics deals with daily-life objects and ordinary phenomena**,** it is consistent with the common sense of our daily experiences, and is easily acceptable and understandable. It was developed earlier before the rise of quantum theory and theory of relativity, so that namely "Classical".

Modern physics deals with microscopic and extremely high speed physical systems, where showing "extraordinary" phenomena, which we could not experience in our daily life so that they seem quite bizarre to us, difficult for us to understand, and against our common sense. It had been developed since the beginning of the 20^{th} century, much later than classical physics, so that namely "Modern".

From two clouds to two storms to two revolutions



Theory of relativity

Quantum mechanics

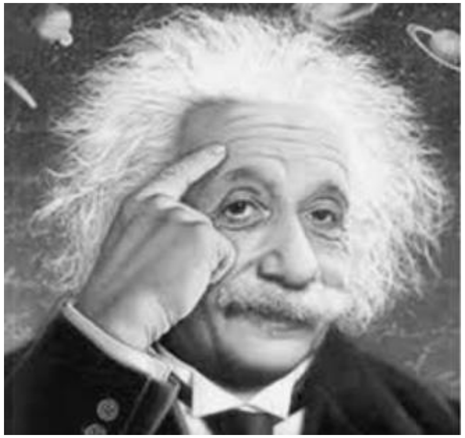
Annus Mirabilis of Physics

Newton in hometown:

1. Calculus
2. Decomposition of light
3. Gravity

Einstein in patent office:

1. 03.18 Photoelectric Effect
2. 04.30 PhD thesis
3. 05.11&12.19 Brownian motion
4. 09.27 $\mathrm{E} = mc^2$
5. 06.30 “On the Eletrodynamics of Moving Bodies”

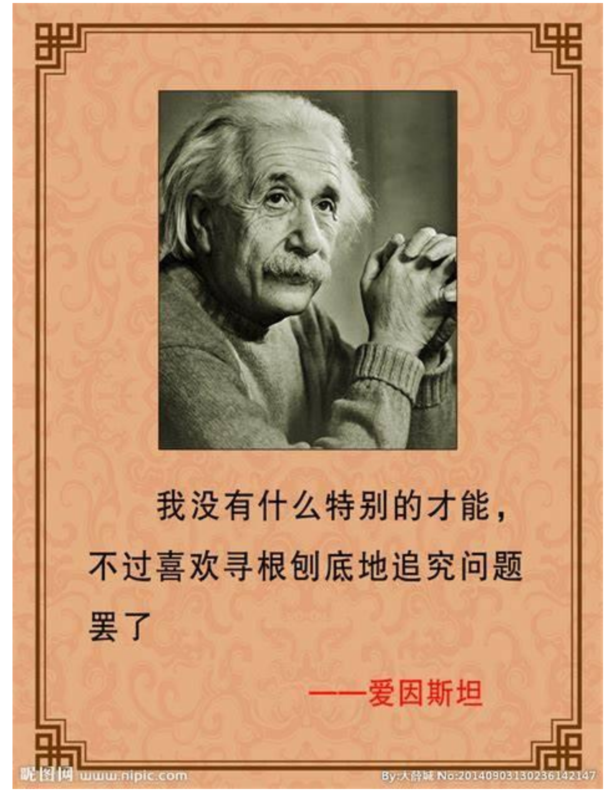


A nickname:
The principle of relativity

Einstein was awarded Nobel Prize for the photoelectric effect.

But photoelectric effect is a “minor” contribution compared to special relativity and general relativity.

Furthermore, Einstein’s predictions about gravitational wave, EPR paradox, quantum gravity etc has set up the next-round Nobel Prize topics.



Albert Einstein (1879-1955) was a German-born theoretical physicist. His intellectual achievements and originality have made the word "Einstein" synonymous with "genius".

Great honor for most scientists who won Nobel Prize.
Great honor for Nobel Prize that Einstein was awarded it.

Chapter 37 Relativity (*Special Relativity*)

Today’s topics (Chapter 37)

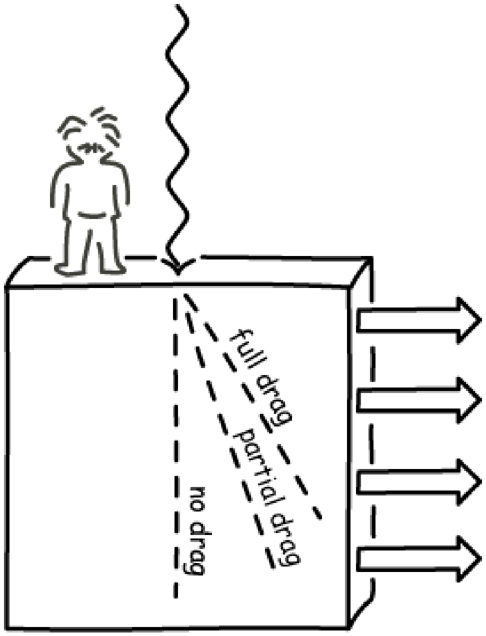
- <u>Time Dilation and Length Contraction</u>
- Lorentz Transformation
- Momentum and Energy
- General Relativity

To save the ether model

Lorentz-Fitzgerald contraction

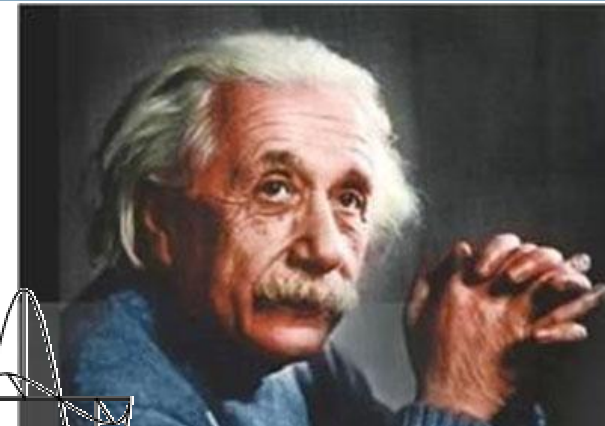
An object contracts along its direction of motion (relative to ether)!

Ether drag

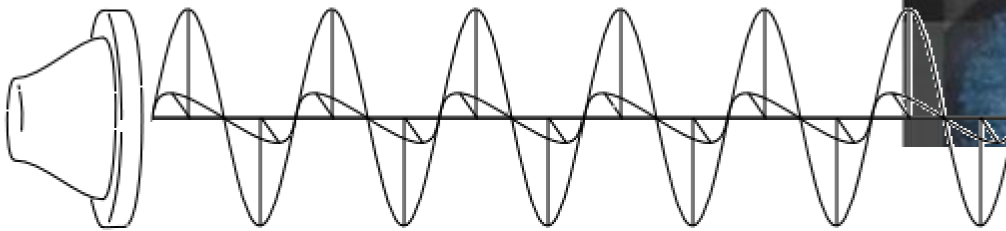


Ether can be dragged by moving objects by various extent.

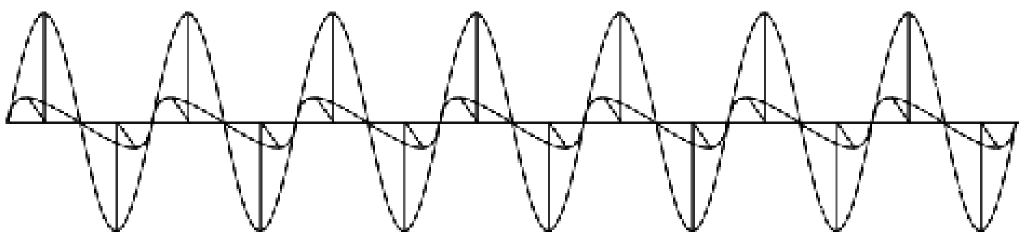
Einstein's thought experiment: chasing light



If he stays still relative to the source:



If he moves with the light at *c*, it will look like (according to the Newtonian mechanics):



?

The two postulates of the special theory of relativity

Galileo said the laws of mechanics were the same in all inertial reference frame (Principle of relativity)

1) Einstein extended this **Principle of relativity: The laws of physics are the same in every inertial frame of reference (no acceleration).**

Example: when you see someone drops a ball in a train moving at a constant velocity, the motion of the ball in the train behaves like as it is dropped at rest

Ohm's law will hold whether you are at rest or your are in a constant moving car

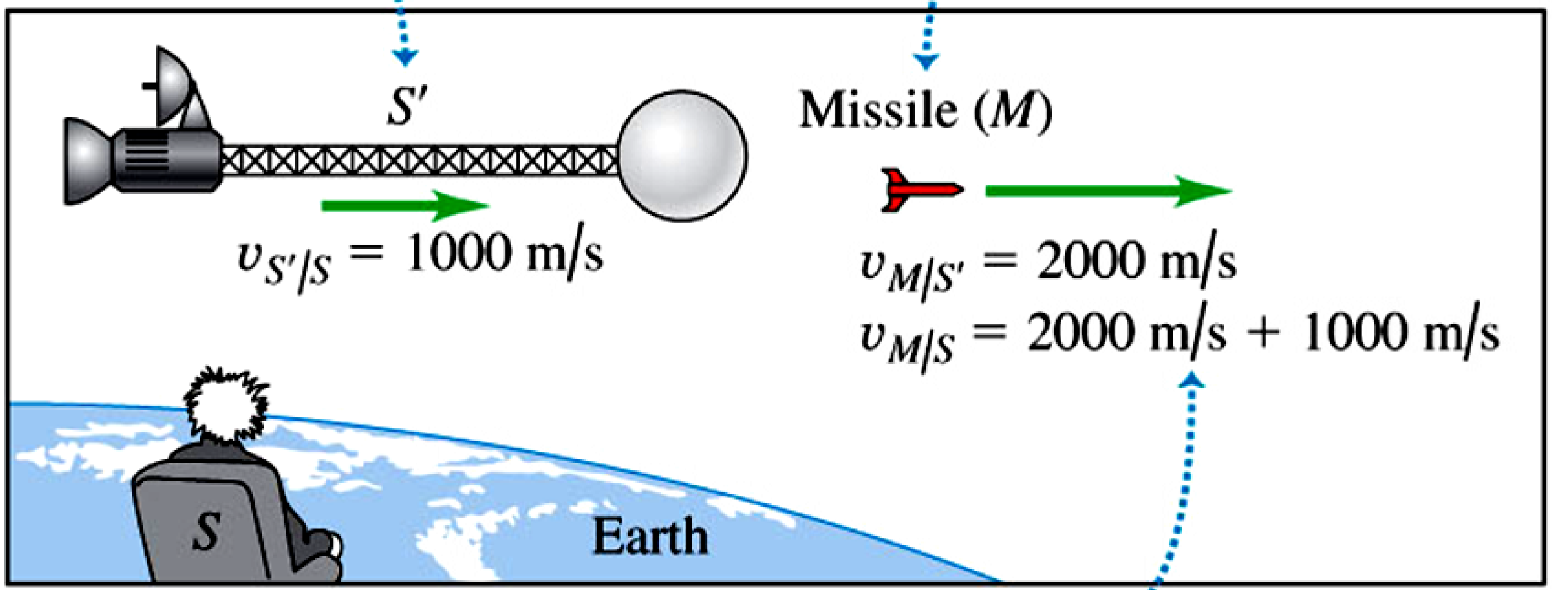
NOT measured values of quantities are the same.
The laws of physics are the same!

2) The postulate of speed of light: The speed of light in vacuum is the same in all inertial frames of reference and is independent of the motion of the source.

The two postulates of the special theory of relativity

A spaceship (S') moves with speed $v_{S'/S} = 1000$ m/s relative to an observer on earth (S).

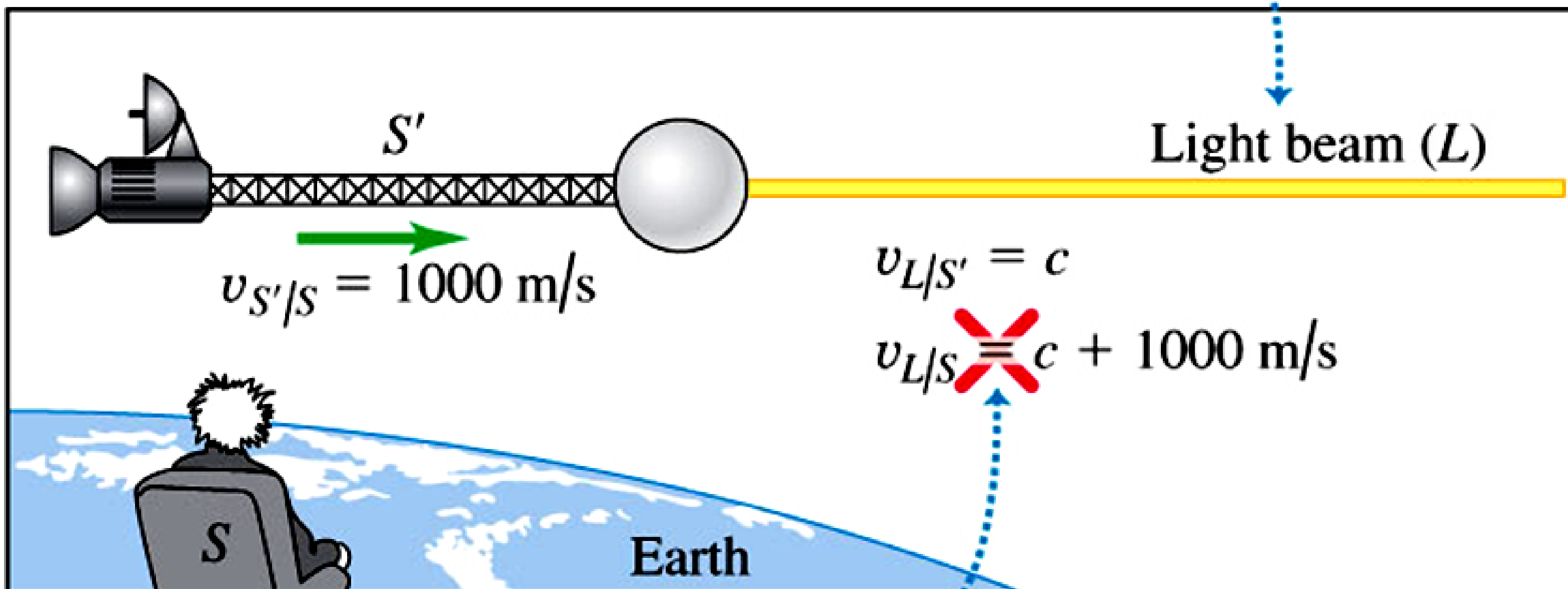
A missile (M) is fired with speed $v_{M/S'} = 2000$ m/s relative to the spaceship.



NEWTONIAN MECHANICS HOLDS: Newtonian mechanics tells us correctly that the missile moves with speed $v_{M/S} = 3000$ m/s relative to the observer on earth.

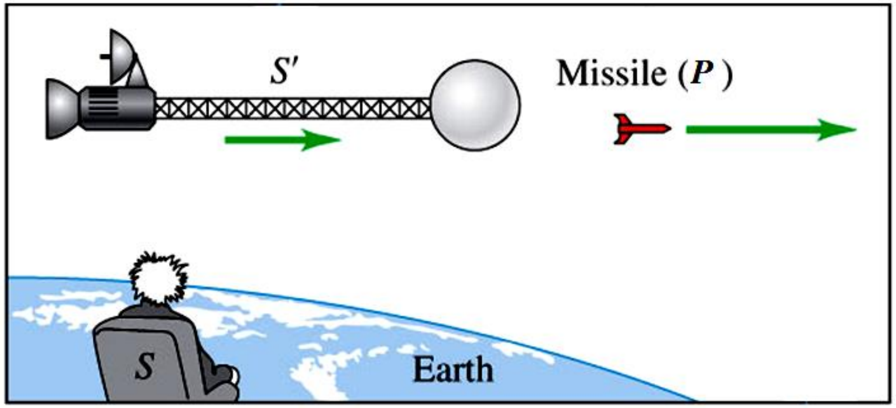
The two postulates of the special theory of relativity

A light beam (L) is emitted from the spaceship at speed c.

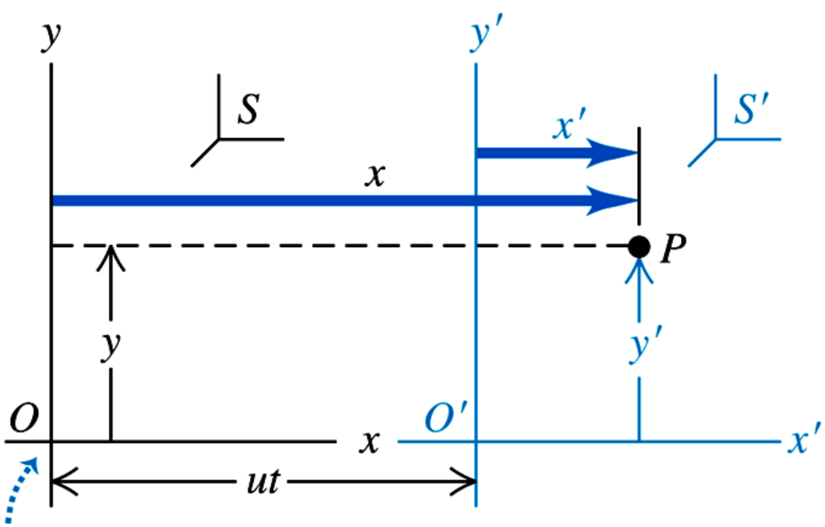


NEWTONIAN MECHANICS FAILS: Newtonian mechanics tells us *incorrectly* that the light moves at a speed greater than c relative to the observer on earth ... which would contradict Einstein's second postulate.

Failure of Newtonian relativity / Galilean coordinate transformation



Frame S' moves relative to frame S with constant velocity u along the common x-x'-axis.



Origins O and O' coincide at time $t = 0 = t'$.

Position of point *P seen by S*

$$x = x' + ut$$

Velocity of point *P seen by S*

$$\frac{dx}{dt} = \frac{dx'}{dt} + u$$

$$v_x = v_x' + u$$

If light is emitted instead of a missile

$$c = c' + u$$

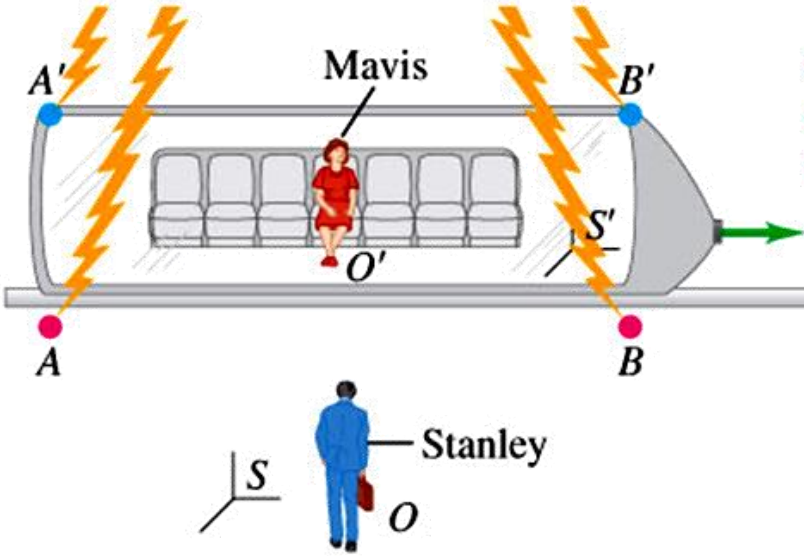
But Einstein's 2nd postulate said $c = c'$

Something must be wrong !!

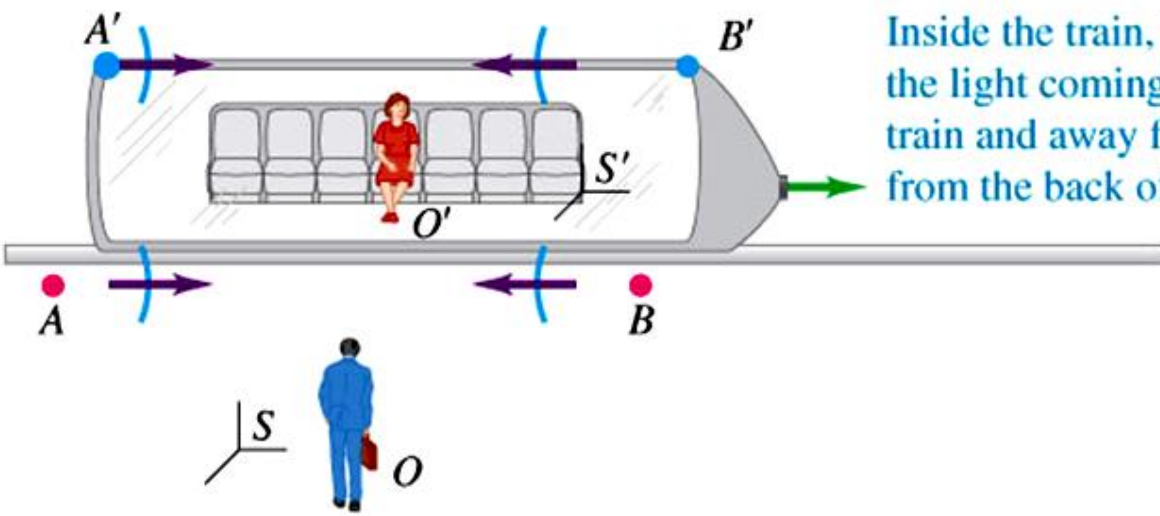
Assumption of S and S' using the same time scale is wrong !

$$v' = dx'/dt', \text{ not as } dx'/dt$$

Relativity of Simultaneity

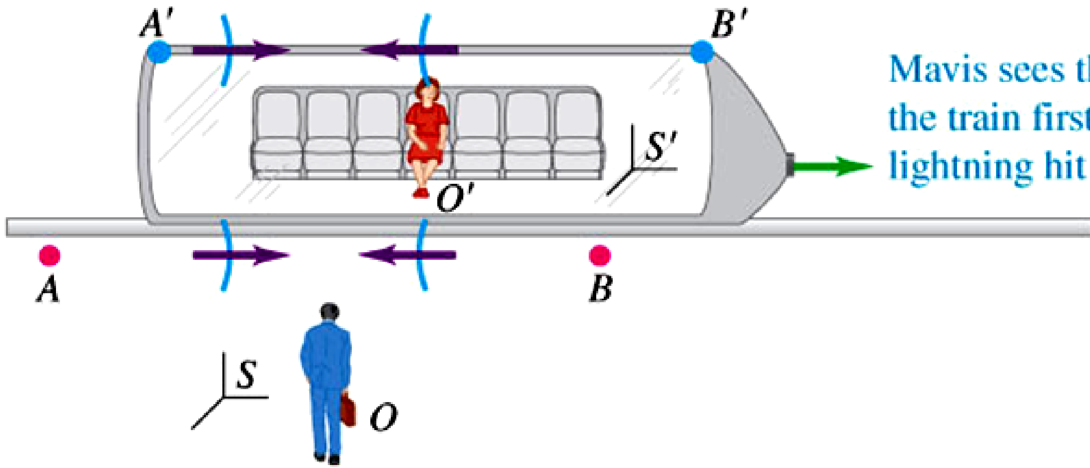


Lightning hits the front and back of a train (points A' and B') and hits the ground at points A and B.

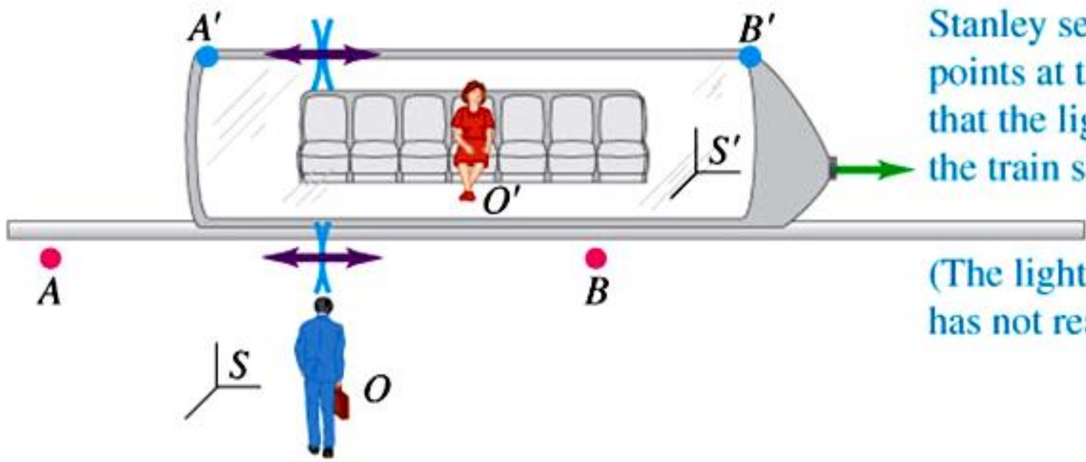


Inside the train, Mavis moves toward the light coming from the front of the train and away from the light coming from the back of the train.

Relativity of Simultaneity



Mavis sees the light from the front of the train first, so she believes that the lightning hit the front of the train first.

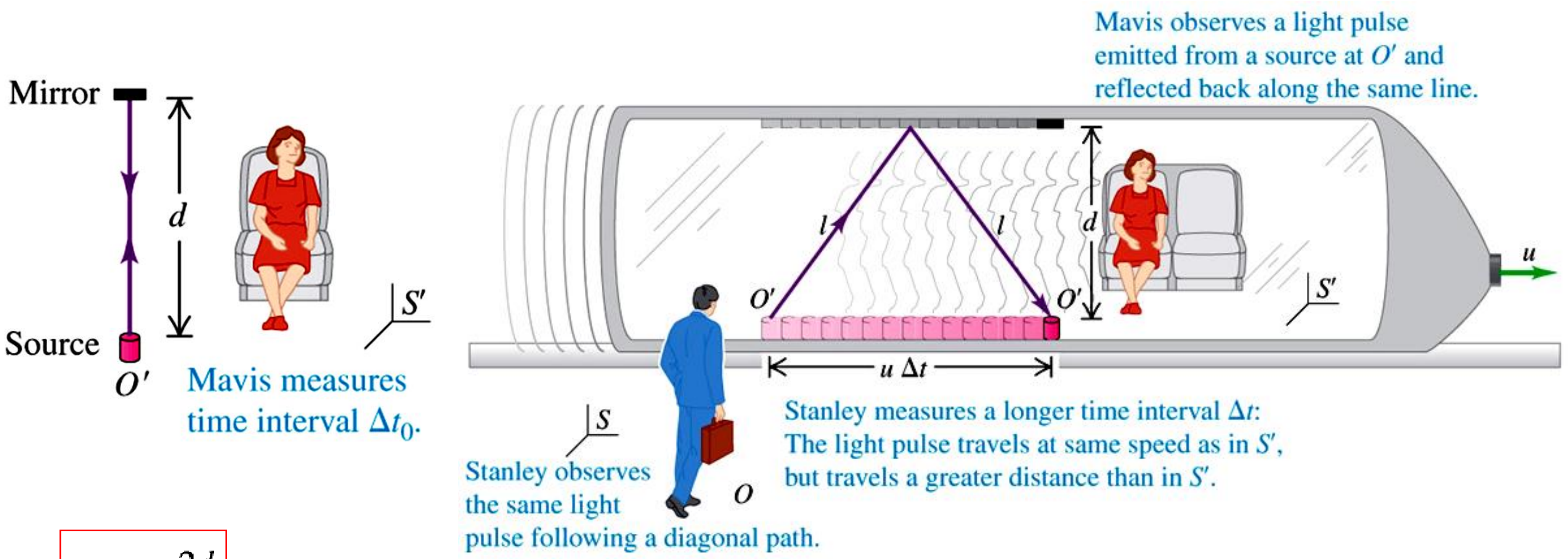


Stanley sees the lightning hit the two points at the same time; so he believes that the lightning hit the two ends of the train simultaneously.

(The light from the rear of the train has not reached Mavis yet.)

Simultaneity is not an absolute concept!
Both are right in their own frame of reference!

Relativity of time intervals



$$\boxed{\Delta t_0 = \frac{2d}{c}}$$

$$l = \sqrt{d^2 + \left(\frac{u\,\Delta t}{2}\right)^2}$$

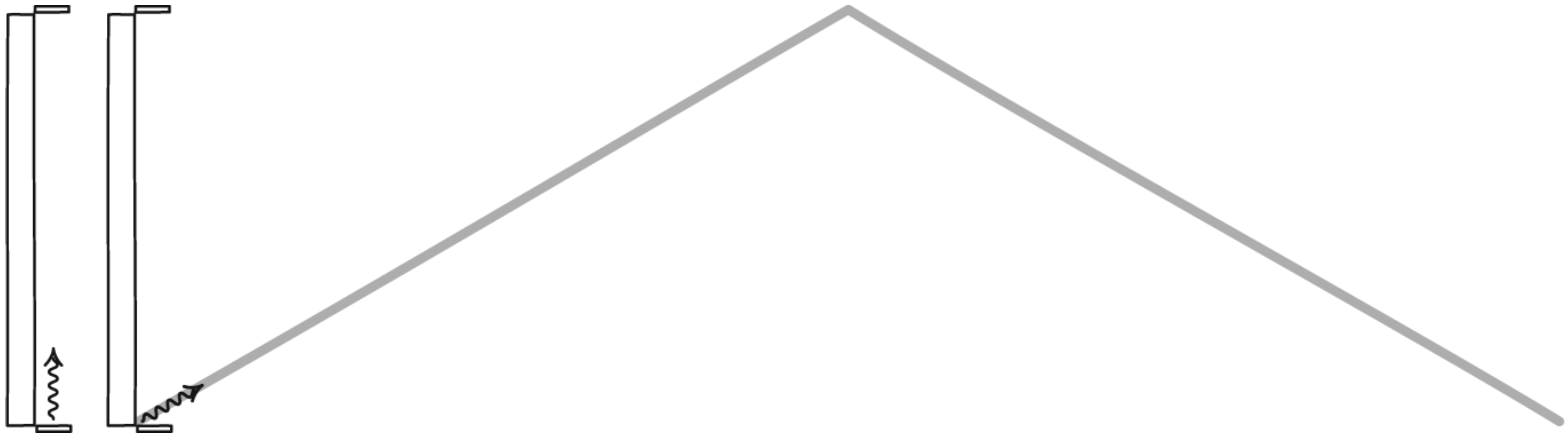
$$\boxed{\Delta t = \frac{2l}{c} = \frac{2}{c}\sqrt{d^2 + \left(\frac{u\,\Delta t}{2}\right)^2}}$$

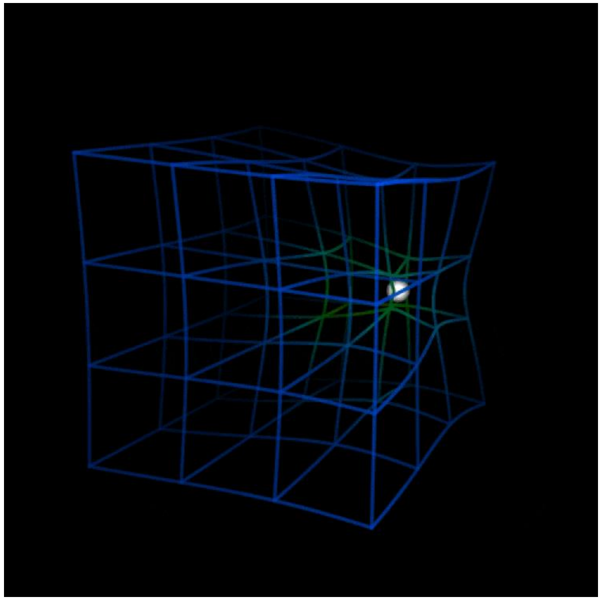
If we combine the 2 equations :

$$\boxed{\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}}}$$

Time dilation

Time dilation (时间膨胀)





Time dilation

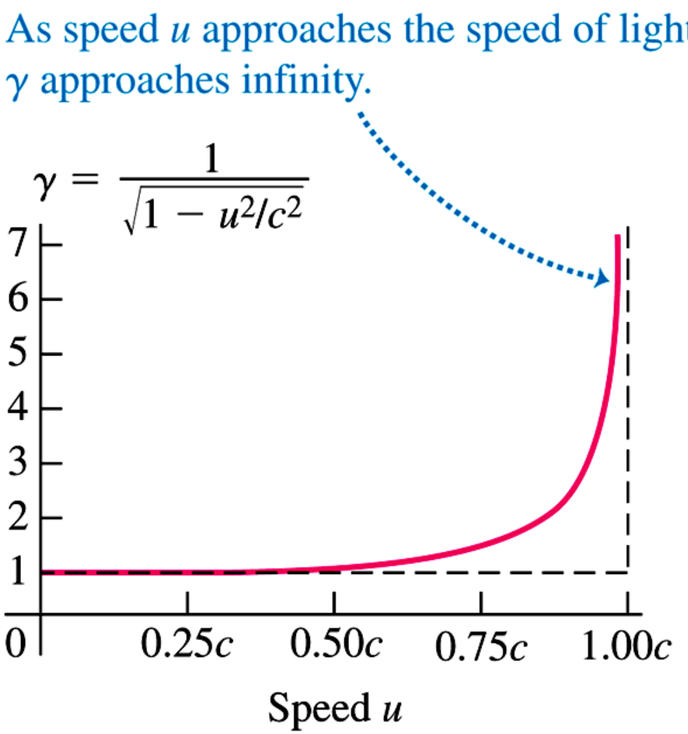
$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}}$$

$$\Delta t = \gamma \, \Delta t_0$$

γ: The Lorentz factor

$$\boxed{\gamma = \frac{1}{\sqrt{1 - u^2/c^2}}}$$

γ is very often used in the theory of relativity

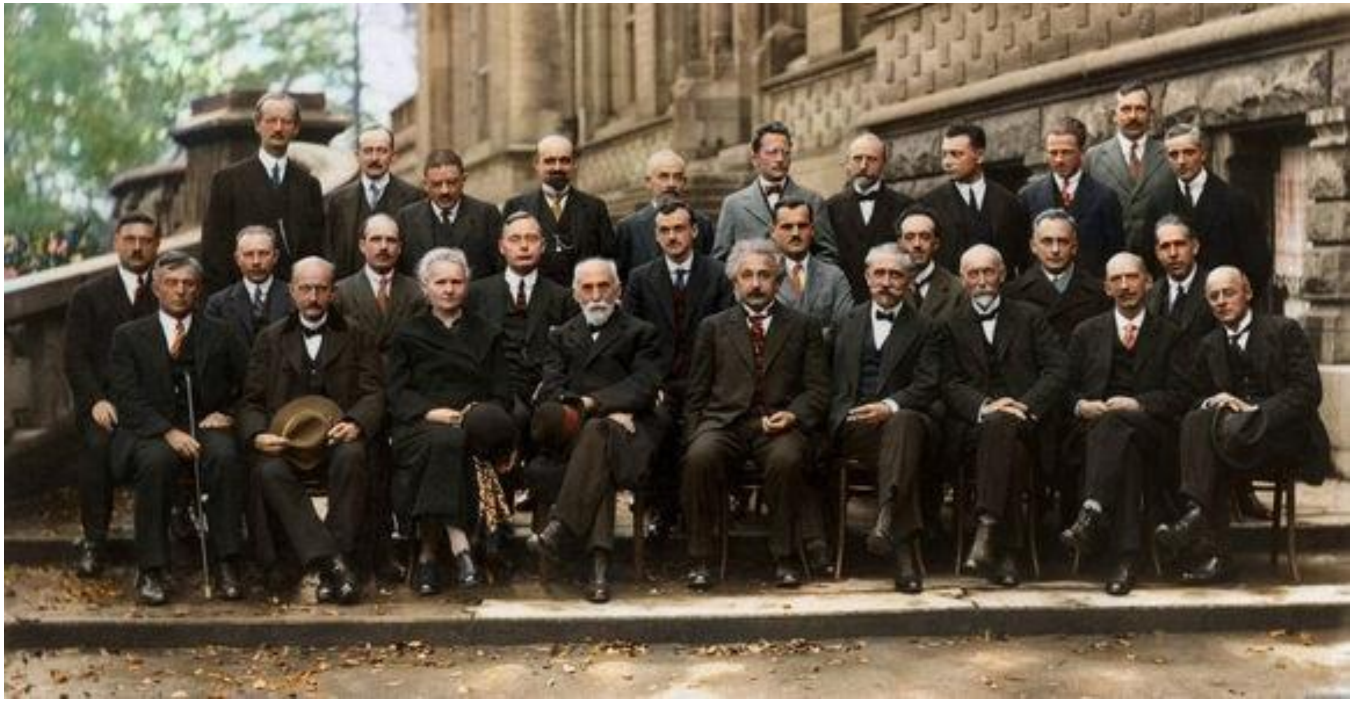


Δt_0 is called the ***proper time***, the shortest duration of an event that any observer can measure (measured in the same inertial frame where the event happens).

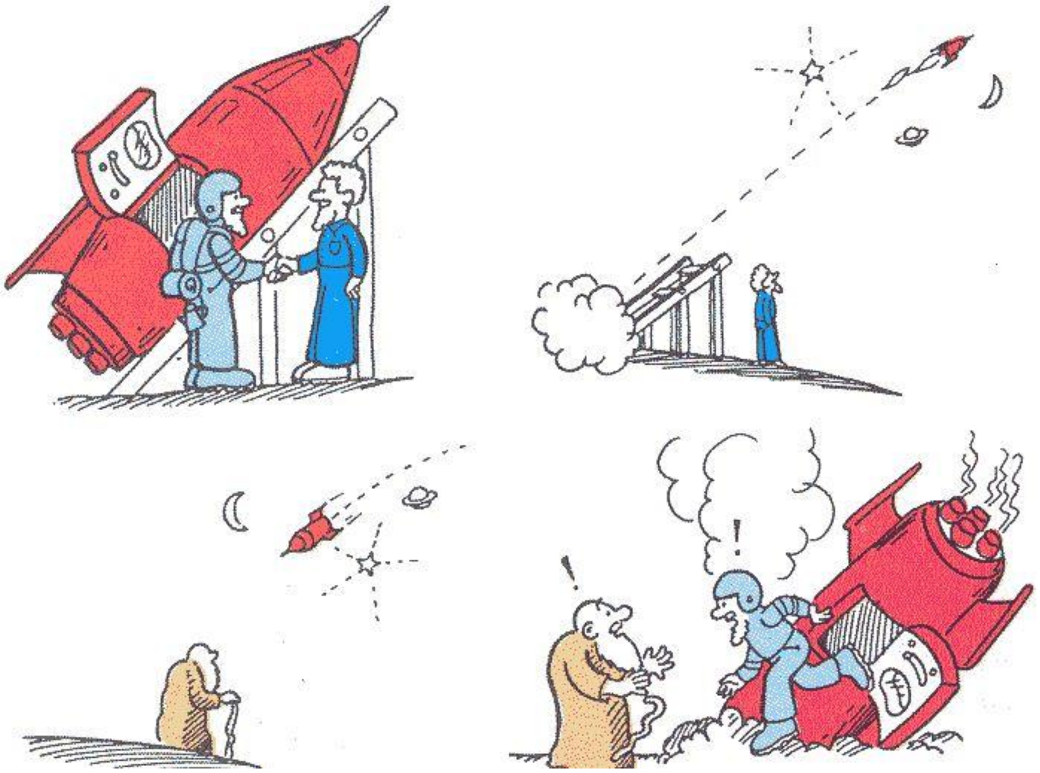
Why called "Lorentz transformation"?

Lorentz transformation was to explain Michelson-Morley experiment. However, Lorentz didn't want to go too much to relate space and time into one entity.

Hendrik Antoon Lorentz (1853 –1928) was a Dutch physicist. 1902 Nobel Prize winner for the discovery of Zeeman effect. Derived the transformation equations underpinning Albert Einstein's theory of special relativity.



Twin Paradox （双生子佯谬）



1911 Paul Langevin

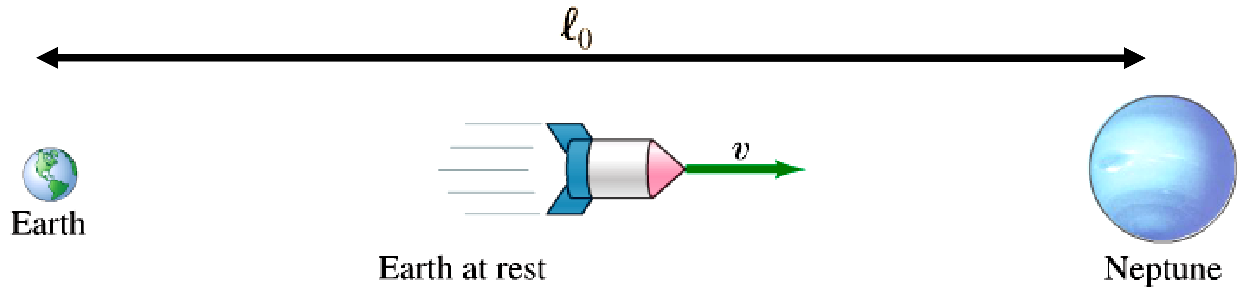
Ground and Sky are twins. Sky flied away at 0.75*c*. Ground waited for Sky on the earth for 30 years. Sky then headed back to the earth at 0.75*c*. After another 30 years on the earth, Sky met with Ground.

Who is younger?

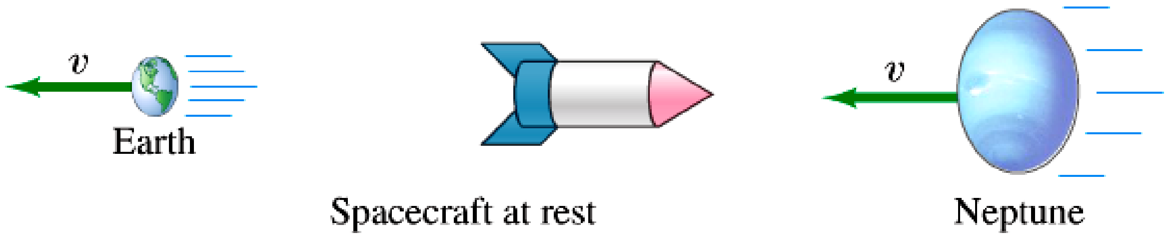
For Ground, $\Delta t = 60$. For Sky, $\Delta t' = \gamma \Delta t \approx 40$. So Sky is younger.

Wait! If in Sky's frame, Ground looks like flying away from Sky. So Sky will think Ground's time slows down! Who the hell is younger then?!

Length contraction



$$\Delta t = \frac{\ell_0}{v}$$ Measured on earth



If I am in the spacecraft, the earth is leaving me at a speed of v and Neptune is coming into me at v

proper time: $\Delta t_0 = \Delta t \sqrt{1 - v^2/c^2} = \Delta t/\gamma$

the length I experience : $\ell = v\,\Delta t_0 = v\,\Delta t\sqrt{1 - v^2/c^2} = \ell_0\sqrt{1 - v^2/c^2}$

the length of an object moving relative to an observer is measured to be shorter along its direction of motion than when it is at rest.

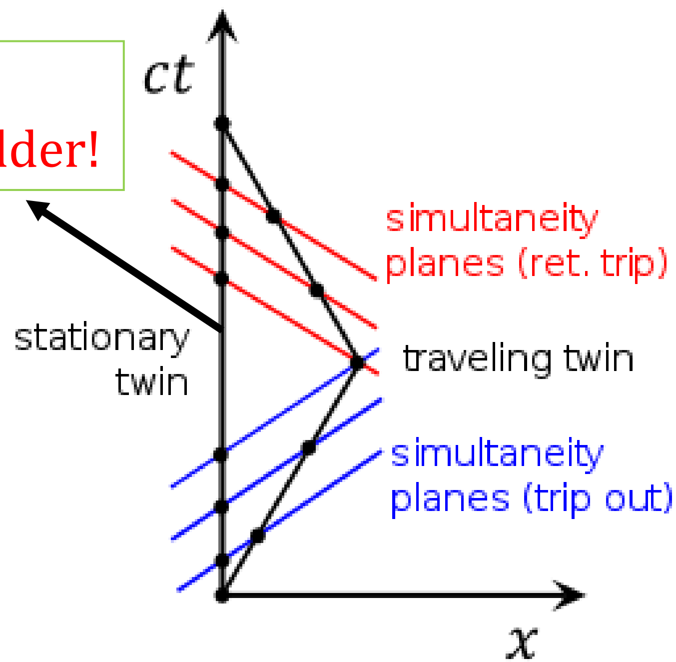
Twin Paradox （双生子佯谬）

The key to solve Twin Paradox is, there must be acceleration and deceleration process when the spaceship turned back!
But, we don't need general relativity to solve the paradox! General relativity is only needed when one takes gravity into account.

Broken line (折线) is longer than straight line! So the time for Sky is longer than that of Ground!

World line (世界线) in a 4-dimensional timespace.
Here we only need *x* and *t*.

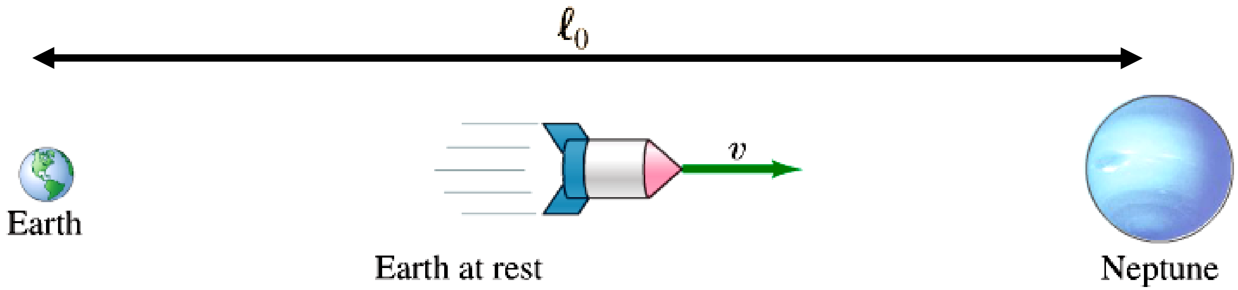
Here is why Ground gets older!



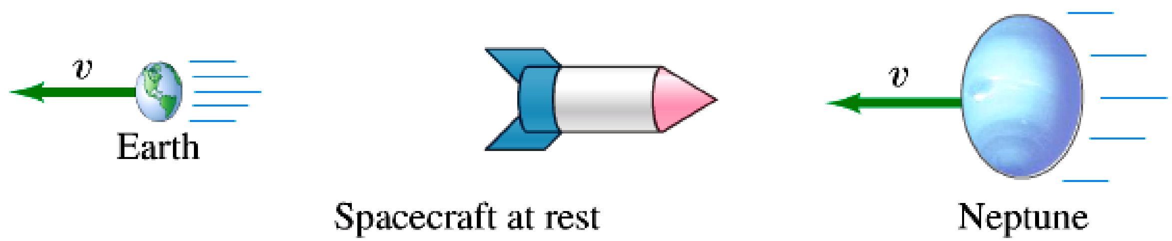
This is Minkowski space (time is real, space is imaginary), totally different from Euclidean space

$$d\tau^2 = dt^2 - dx^2$$

Relativity of Length



$$\Delta t = \frac{\ell_0}{v}$$ Measured on earth (not proper time here, why?)



If I am in the spacecraft, the earth is leaving me at a speed of ***v*** and Neptune is coming into me at ***v*.** the length I experience :

$$\ell = v\,\Delta t_0 = v\,\Delta t\sqrt{1 - v^2/c^2} = \ell_0\sqrt{1 - v^2/c^2}$$

The length l_0 of an object measured in the rest frame is its **proper length.** Measurements from any reference frame that is in relative motion *parallel* to that length are always less than the proper length.

Length contraction

If this picture is inside a spacecraft which travels at 0.90c

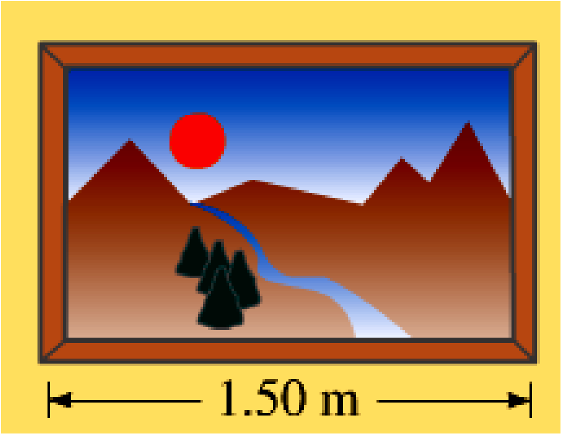
The dimension of the picture seen on earth :

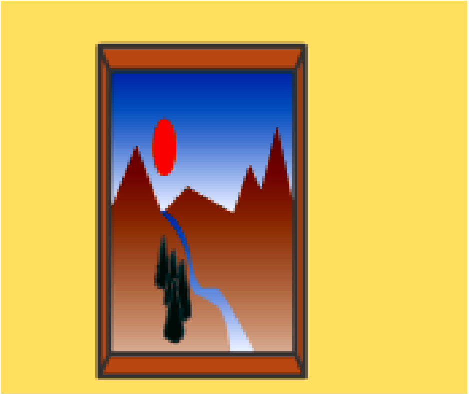
$$\ell = \ell_0\sqrt{1 - \frac{v^2}{c^2}}$$

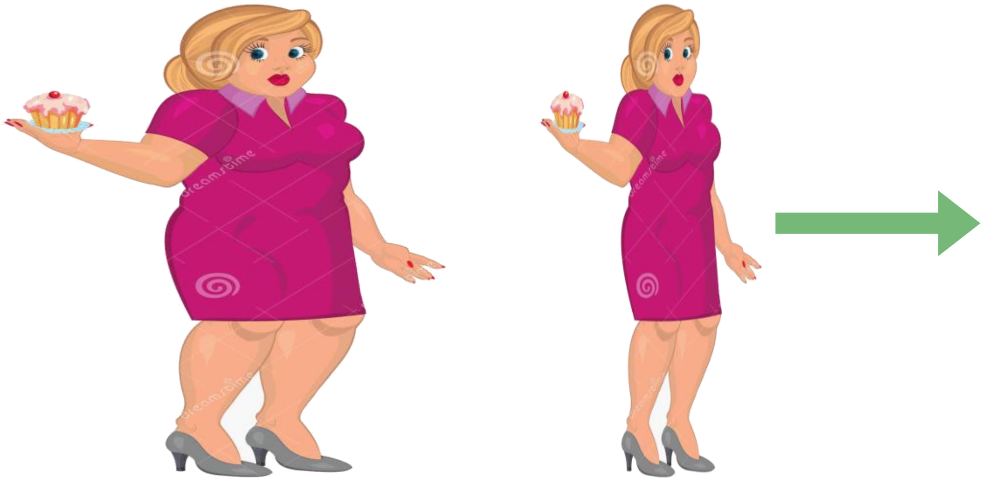
$$= (1.50\,\mathrm{m})\sqrt{1 - (0.90)^2}$$

$$= 0.65\,\mathrm{m}$$

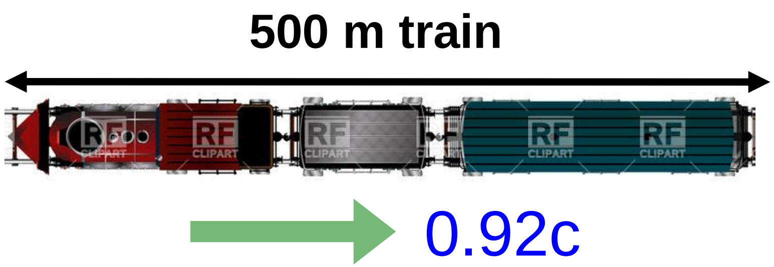
Lorentz-Fitzgerald contraction

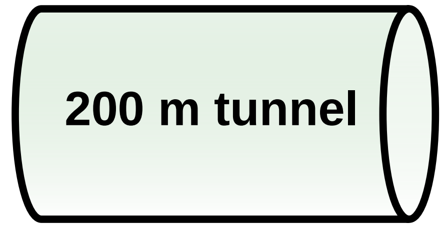






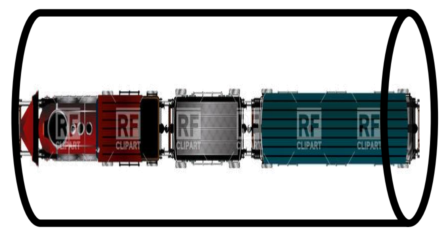
Length contraction





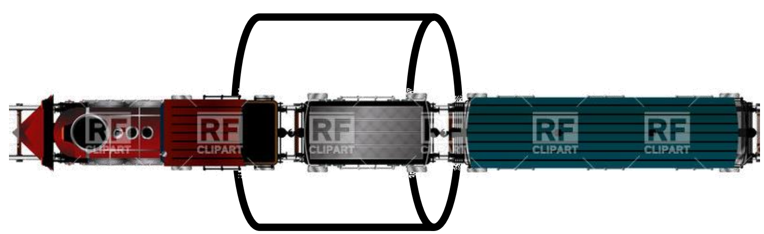
If I am standing outside seeing the train travel at 0.92 c going into a tunnel at rest, I will see the length of the train becomes :

$$\ell = \ell_0\sqrt{1-\frac{v^2}{c^2}} = 200 \text{ m}$$



If I am inside the train, I will see the tunnel becomes :

$$\ell = \ell_0\sqrt{1-\frac{v^2}{c^2}} = 80 \text{ m}$$



The apparent contradiction is explained by the relativity of simultaneity.

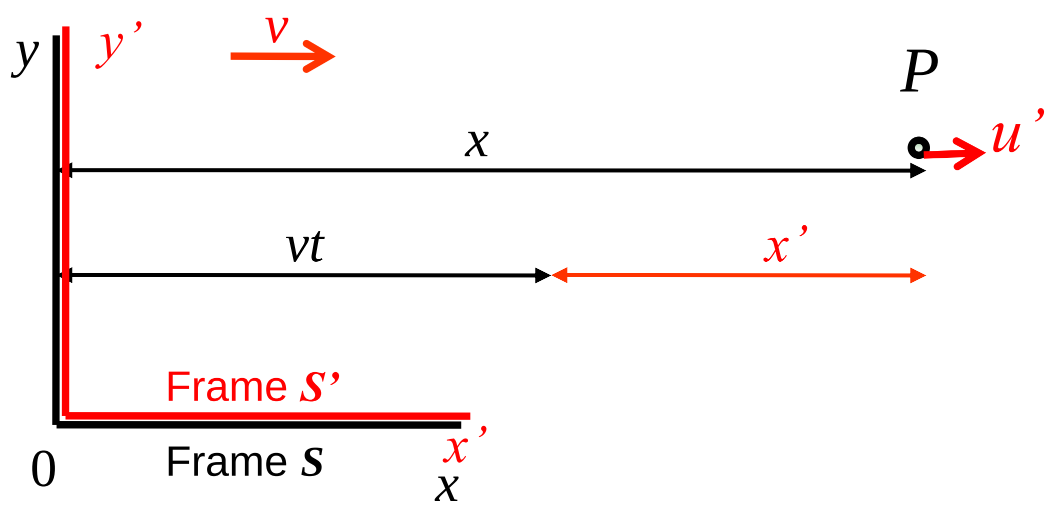
What happens if there is a hard wall on the right end of the tunnel?
Can the train be fit into the tunnel?

Today's topics (Chapter 37)

- Time Dilation and Length Contraction
- <u>Lorentz Transformation</u>
- Momentum and Energy
- General Relativity

Galilean transformation vs. Lorentz transformation

In general, we need to consider coordinate transformation between two reference frames in order to relate events observed in them.



at time t'

Galilean transformation equations

(transform the events in **Frame S'** to **Frame S**)

$$x = x' + vt' \qquad t = t'$$

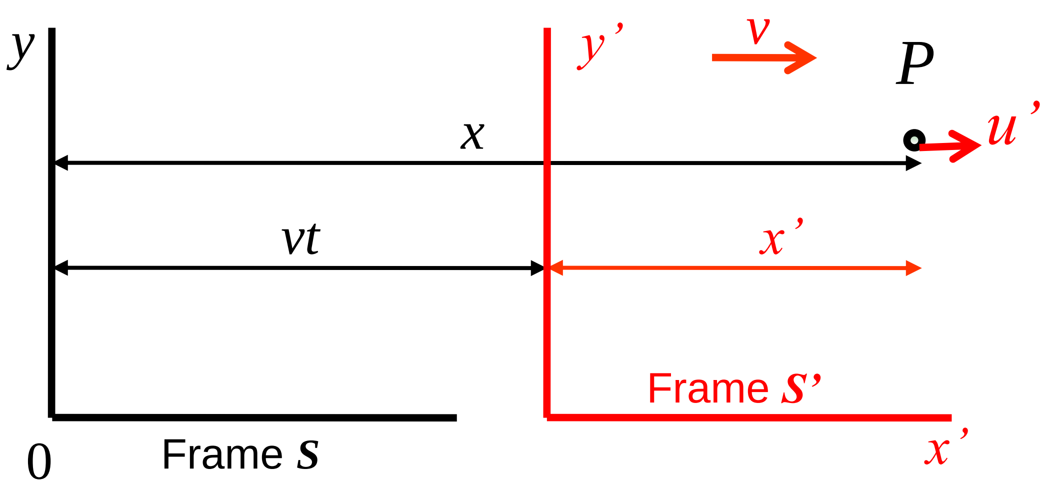
Galilean velocity transformation equations.

(transform the events in **Frame S'** to **Frame S**)

$$u = \frac{dx}{dt} = \frac{d(x' + vt')}{dt'} = u' + v$$

Galilean transformation vs. Lorentz transformation

Lorentz transformation:



$$x = x' + vt' \qquad x' = x - vt$$

$$x = \gamma(x' + vt') \qquad x' = \gamma(x - vt)$$

$$\boxed{\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}}$$

Galilean transformation vs. Lorentz transformation

To relate ***t*** and ***t'***, we combine

$$x' = \gamma(x - vt) \qquad x = \gamma(x' + vt')$$

$$x' = \gamma(x - vt) = \gamma(\gamma[x' + vt'] - vt)$$

We solve for t and find

$$\boxed{t = \gamma\left(t' + \frac{vx'}{c^2}\right)}$$

We solve for t' and find

$$\boxed{t' = \gamma\left(t - \frac{vx}{c^2}\right)}$$

To relate ***u*** and ***u'***, we differentiate $x = \gamma(x' + vt')$

$$u = \frac{dx}{dt} = \frac{d}{dt}[\gamma(x' + vt')] = \frac{d}{dt'}[\gamma(x' + vt')]\frac{dt'}{dt}$$

since $dx'/dt' = u' \quad dt'/dt = 1/(dt/dt')$

$$u = \frac{[\gamma(u' + v)]}{[\gamma(1 + vu'/c^2)]} \boxed{= \frac{u' + v}{1 + vu'/c^2}}$$

Lorentz transformation of velocity

Galilean transformation vs. Lorentz transformation

Galilean transformation:

$$x = x' + vt'$$

$$t = t'$$

$$u = u' + v$$

$$a' = a$$

Lorentz transformations:

$$\boxed{\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}}$$

$$x = \gamma(x' + vt')$$

$$t = \gamma\left(t' + \frac{vx'}{c^2}\right)$$

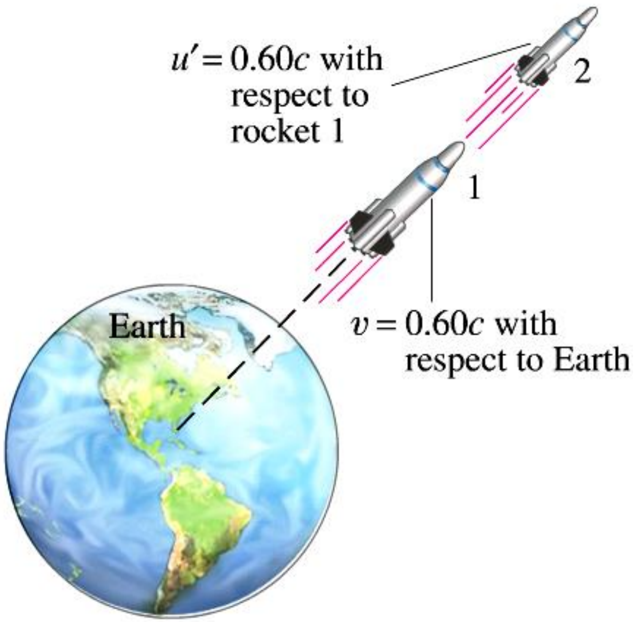
$$u = \frac{u' + v}{1 + vu'/c^2}$$

$$a' = \frac{a}{\gamma^3(1 - vu_x/c^2)^3}$$

Example : Adding velocities

Consider Earth as reference frame S, and rocket 1 as reference frame S'. Rocket 2 moves with speed u' = 0.60c with respect to rocket 1. Rocket 1 has speed *v* = 0.60c with respect to Earth.

What is the speed of rocket 2 with respect to Earth ?



$$u = \frac{u'_x + v}{1 + vu'_x/c^2}$$

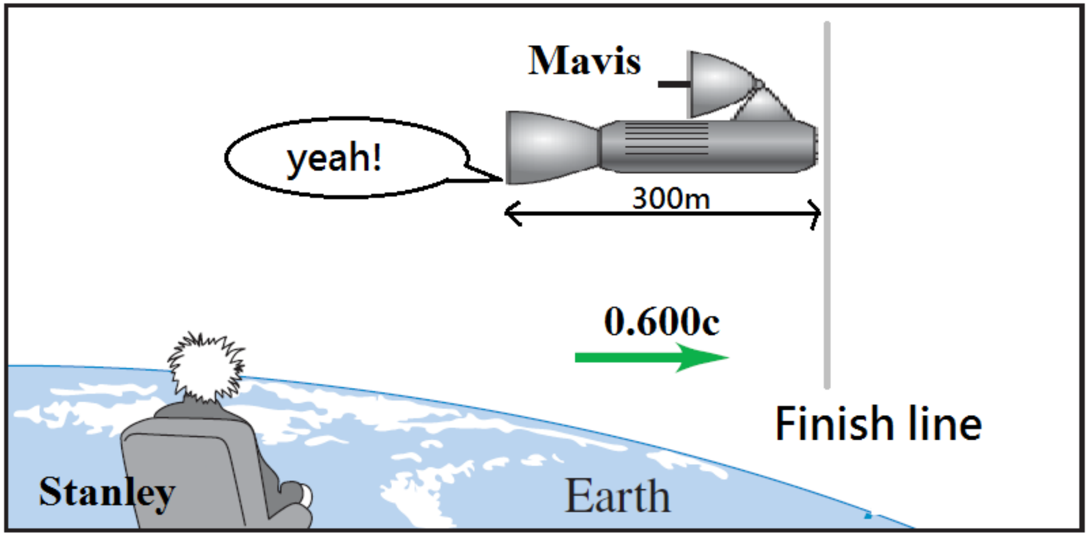
Lorentz velocity transformations

$$= \frac{0.60c + 0.60c}{1 + \frac{(0.60c)(0.60c)}{c^2}}$$

$$= \frac{1.20c}{1.36}$$

$$= 0.88c$$

Example :



At a spaceship race, ***Mavis*** sent a message at the back of her ship (event 2) at the instant when the front of her ship passes the finish line (event 1). (in her reference frame *S'*)

How does Stanley measure event 1 & 2 to occur?

For simplicity, we set origin of *S* (for ***Stanley***) at the finish line and origin of *S'* (for ***Mavis***) at the front of her ship. The two origins coincide at $t = 0 = t'$.

∴ ***Stanley*** and ***Mavis*** measure event 1 to be at $x = 0 = x'$ and $t = 0 = t'$

For ***Mavis***, event 2 measured to be at $x' = -300$ and at $t' = 0$

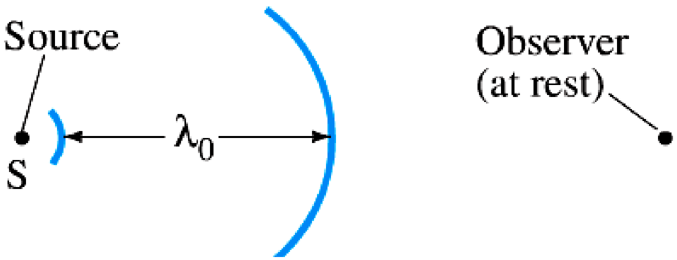
For ***Stanley***, $\boxed{x = \gamma(x' + ut') \quad \text{and} \quad t = \gamma(t' + ux'/c^2)}$

$$\gamma = 1.25 \text{ for } u = 0.600c = 1.80 \times 10^8 \text{ m/s} \qquad x' = -300 \text{ m}, \; t' = 0$$

He measured event 2 $\quad x = -375 \text{ m at } t = -7.50 \times 10^{-7} \text{ s} = -0.750 \; \mu\text{s}$

Stanley saw the “yeah!” -0.75 μs before event 1 at *t = 0, what’s wrong?*

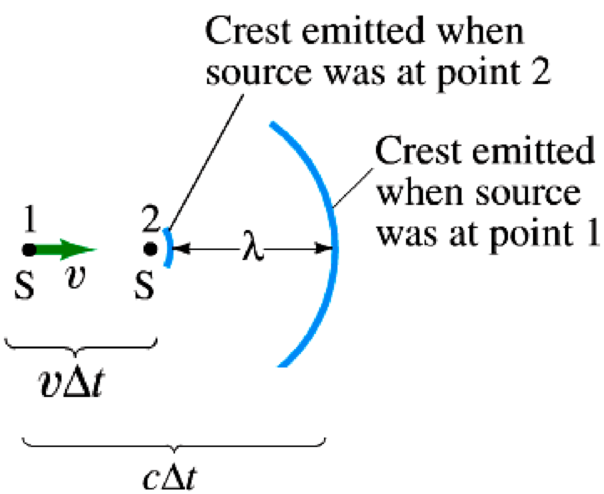
Doppler shift for light



$$\lambda = c\,\Delta t - v\,\Delta t$$

$$\Delta t = \Delta t_0 / \sqrt{1 - v^2/c^2}$$

Time in the source stationary reference frame



$$\lambda = (c - v)\,\Delta t$$

$$= (c - v)\frac{\Delta t_0}{\sqrt{1 - v^2/c^2}}$$

$$\boxed{\Delta t_0 = \frac{\lambda_0}{c}}$$

$$= \frac{(c - v)}{\sqrt{c^2 - v^2}}\lambda_0$$

$$\boxed{\lambda = \lambda_0\sqrt{\frac{c - v}{c + v}}}$$

(source and observer moving toward each other)

$$\boxed{\lambda = \lambda_0\sqrt{\frac{c + v}{c - v}}}$$

(source and observer moving away from each other)

Example of Doppler effect

A driver claims that he did not go through a red light because the light was Doppler shifted and appeared green. What would be the speed of the car to see this effect?

red light ($\lambda_0 \approx 650$ nm)

green light ($\lambda \approx 500$ nm)

$$\lambda = \lambda_0 \sqrt{\frac{c - v}{c + v}}$$

(source and observer moving toward each other)

$$\left(\frac{\lambda}{\lambda_0}\right)^2 = \frac{c - v}{c + v} \qquad (\lambda/\lambda_0)^2 = (500\ \text{nm}/650\ \text{nm})^2$$

We solve for v: $\quad v = c\left[\frac{1 - (\lambda/\lambda_0)^2}{1 + (\lambda/\lambda_0)^2}\right] = 0.26c$

Today's topics (Chapter 37)

- Time Dilation and Length Contraction
- Lorentz Transformation
- <u>Momentum and Energy</u>
- General Relativity

Momentum

Total momentum is NOT conserved for the observers in different inertial frames! Physicists still want to make the conservation law hold, so momentum in relativity is redefined as

$$p = m\frac{\Delta x}{\Delta t_0} = m\frac{\Delta x}{\Delta t}\frac{\Delta t}{\Delta t_0} = m\frac{\Delta x}{\Delta t}\gamma$$

Δx: distance by earth observer;
Δt_0: time with the moving object

Momentum in relativity:

$$\boxed{\vec{p} = \gamma m \vec{v}}$$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

Energy

Energy is related to work:

$$W = \int_{x_1}^{x_2} F dx = \int_{x_1}^{x_2} \frac{dp}{dt} dx = \int_{x_1}^{x_2} \frac{dp}{dv}\frac{dv}{dt} dx = \int_{v_1}^{v_2} \frac{dp}{dv} v dv$$

Relativistically

$$p = \gamma m v = \frac{mv}{\sqrt{1 - v^2/c^2}}$$

$$W = \int_{v_1}^{v_2} \frac{dp}{dv} v dv = \frac{mc^2}{\sqrt{1 - {v_2}^2/c^2}} - \frac{mc^2}{\sqrt{1 - {v_1}^2/c^2}}$$

Energy in relativity:

Try yourself!

$$\boxed{E = \gamma m c^2}$$

Total Energy

Total Energy:

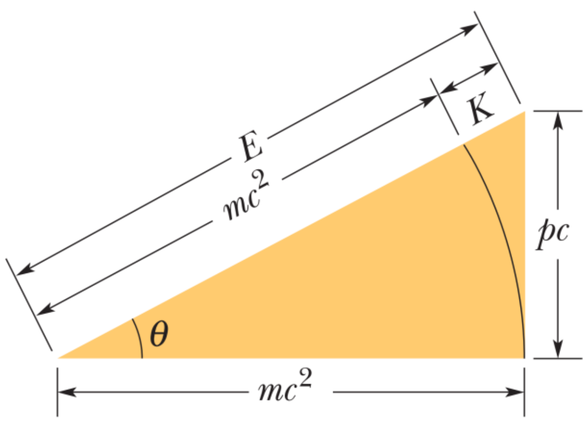
$$E = E_0 + K = mc^2 + K$$

$$\boxed{K = mc^2(\gamma - 1)}$$

E_0: energy of the rest mass; K: kinetic energy

Energy relation:

$$\boxed{E^2 = (pc)^2 + (mc^2)^2}$$



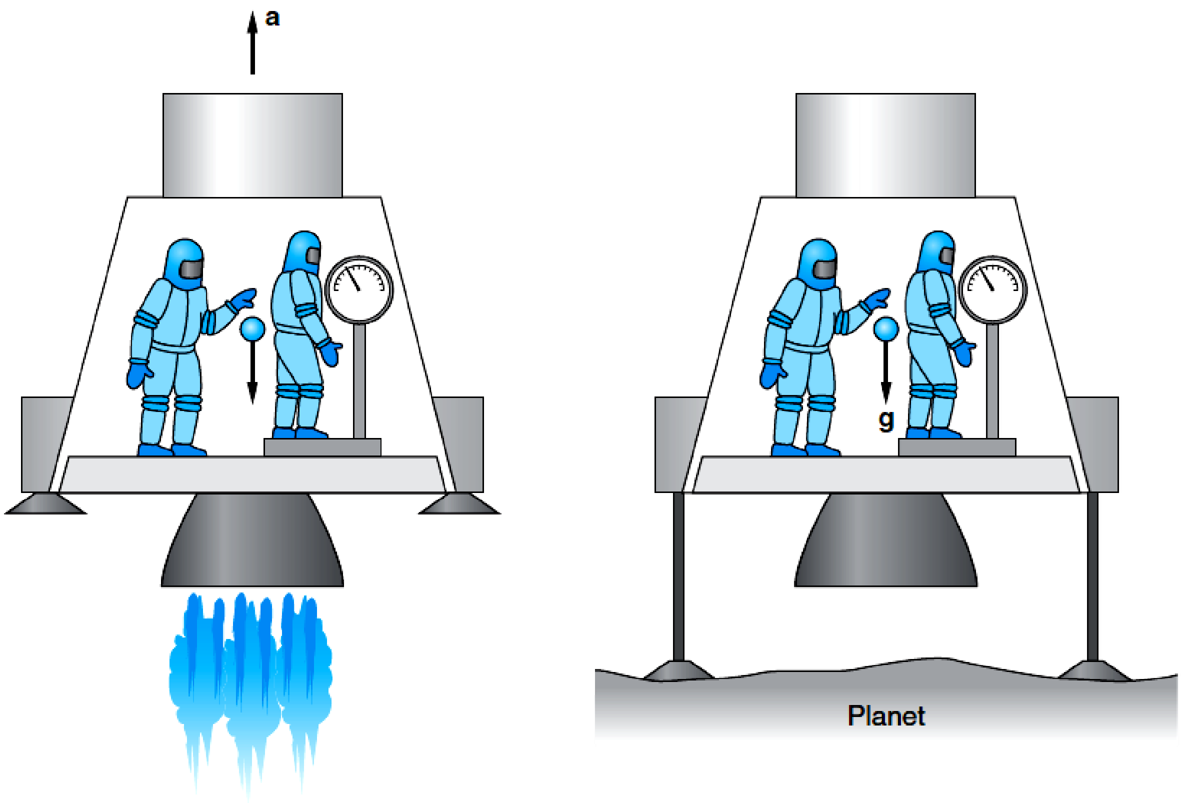
Today's topics (Chapter 37)

- Time Dilation and Length Contraction
- Lorentz Transformation
- Momentum and Energy
- General Relativity

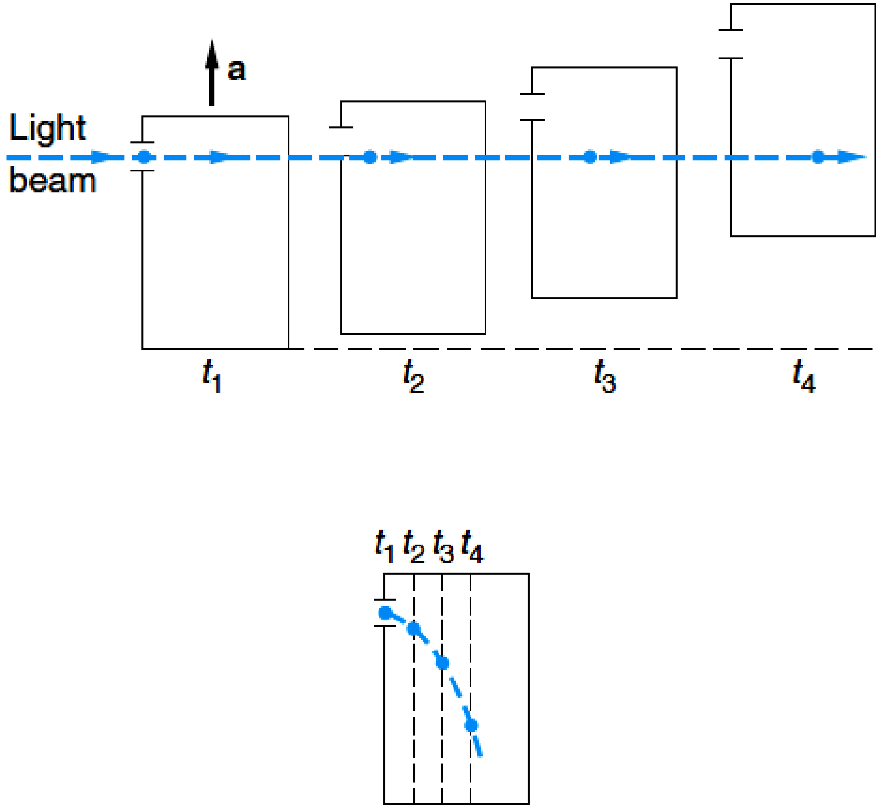
Principle of Equivalence

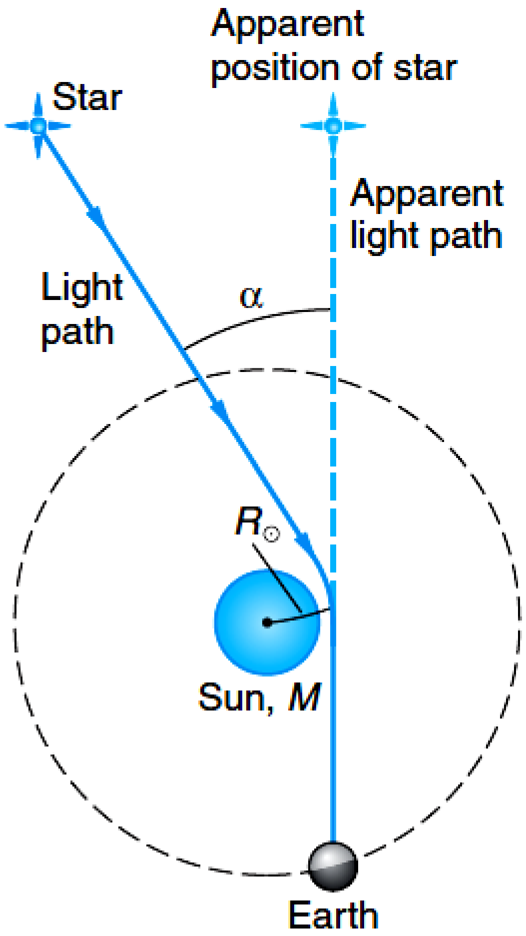
The basis of the general theory of relativity is what we may call Einstein's third postulate, the principle of equivalence, which states:

A homogeneous gravitational field is completely equivalent to a uniformly accelerated reference frame.



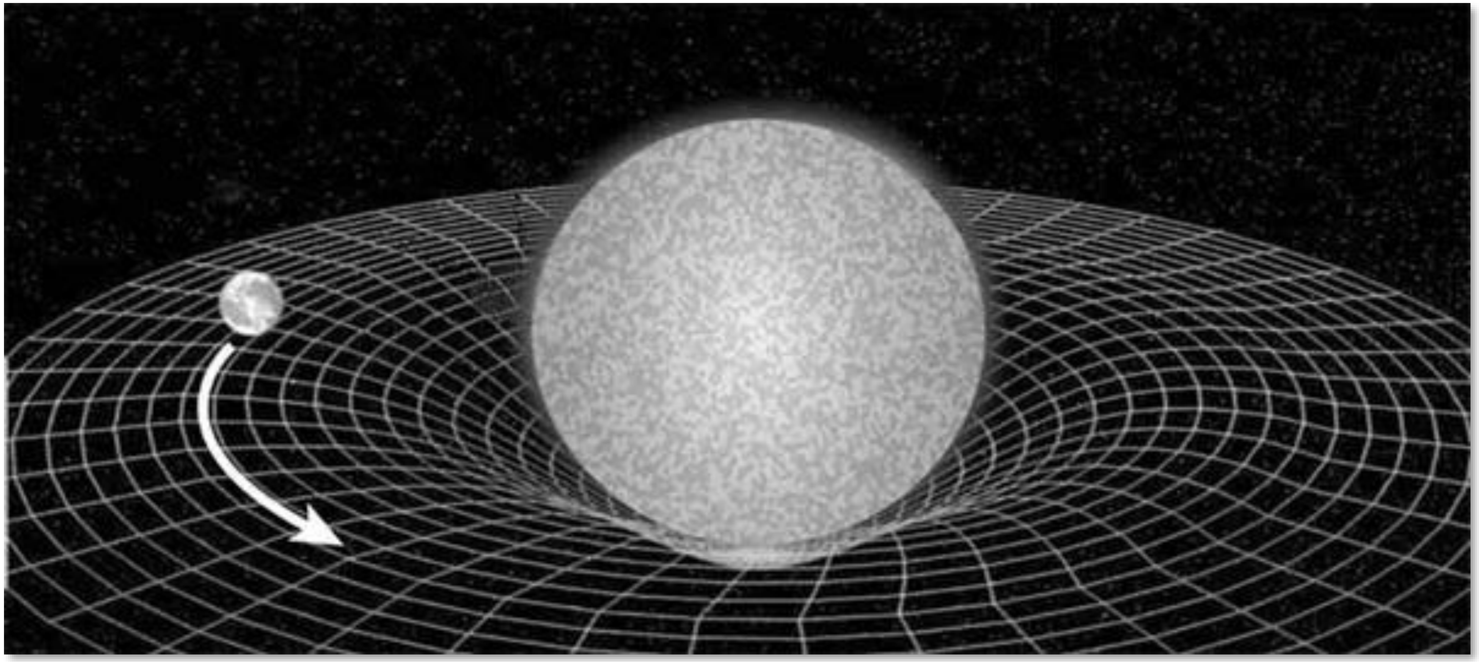
Deflection of light in a Gravitational Field





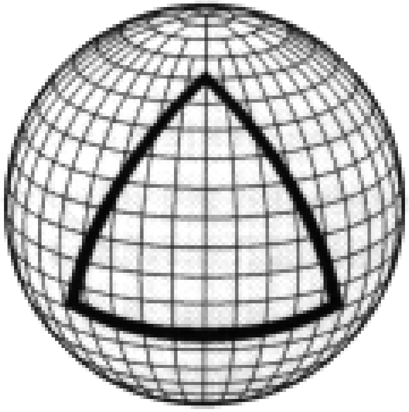
- In 1919, Eddington's expedition confirmed the bending of light in a gravitational field!

There is really no such thing called gravitation…

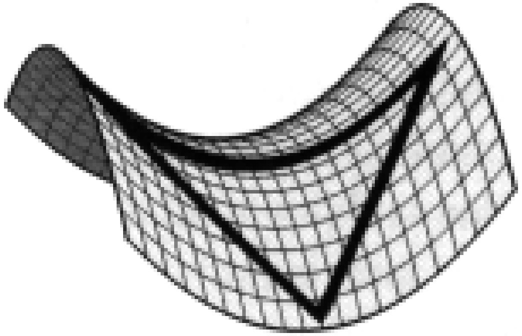


- The essence of general relativity: spacetime becomes curved by huge masses, such as the Sun, and the Earth's trajectory is basically a path of the minimum distance in such a curved spacetime, the so called geodesic. In some sense, gravitation is geometrized in general relativity.

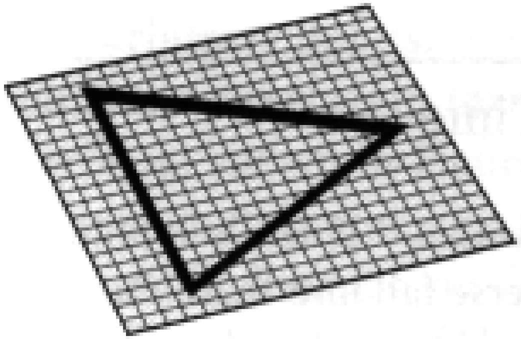
Curved space



Positive Curvature

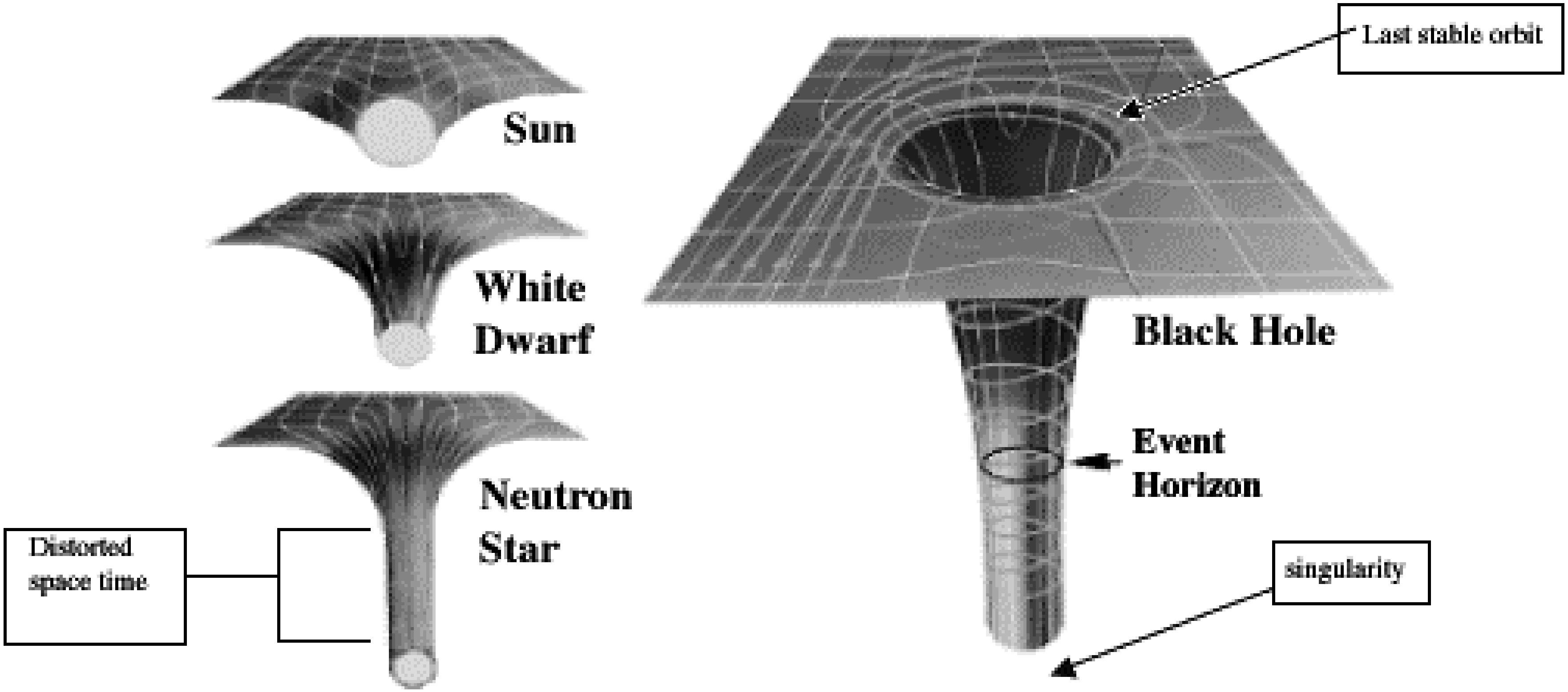


Negative Curvature

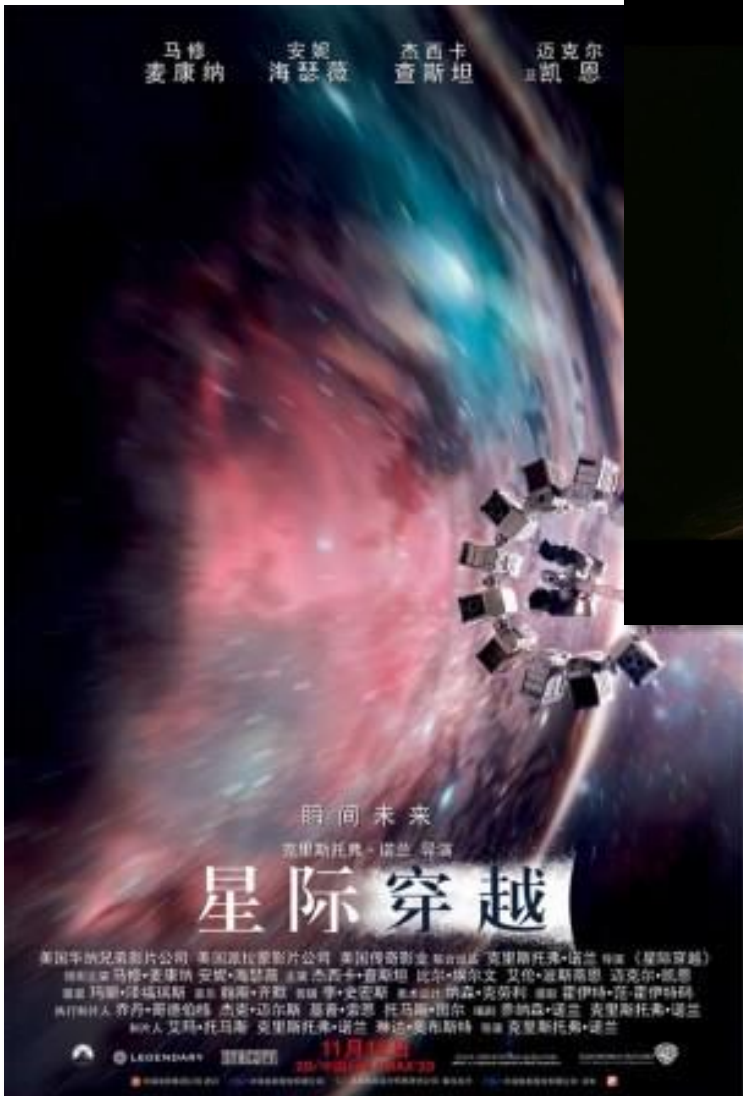


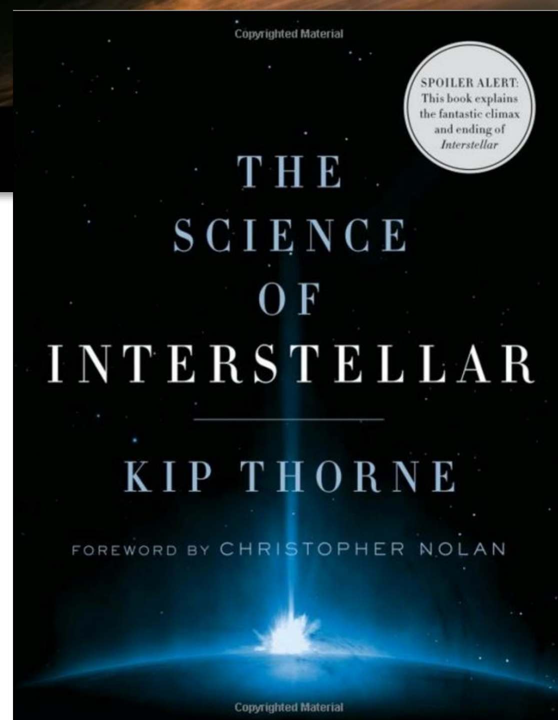
Flat Curvature

Curved space around huge masses



More about black hole and theory of relativity





Detection of gravitational wave

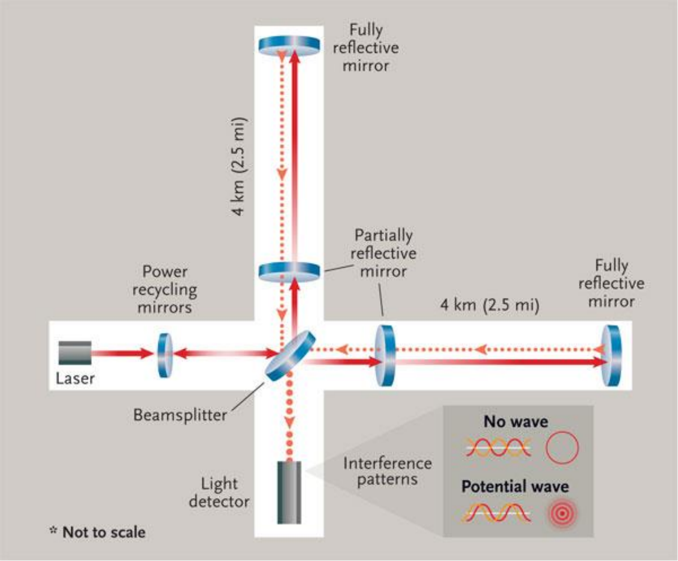
Official announcement: ~~http://www.youtube.com/watch?v=aEPIwEJmZyE~~

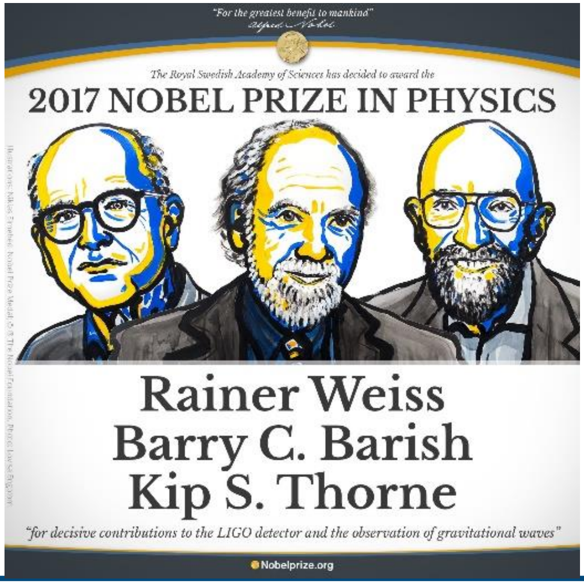
Brian Greene explains GW: ~~http://www.youtube.com/watch?v=s06_jRK939I~~

New Yorker article:
~~http://www.newyorker.com/tech/elements/gravitational-waves-exist-heres-how-scientists-finally-found-them~~

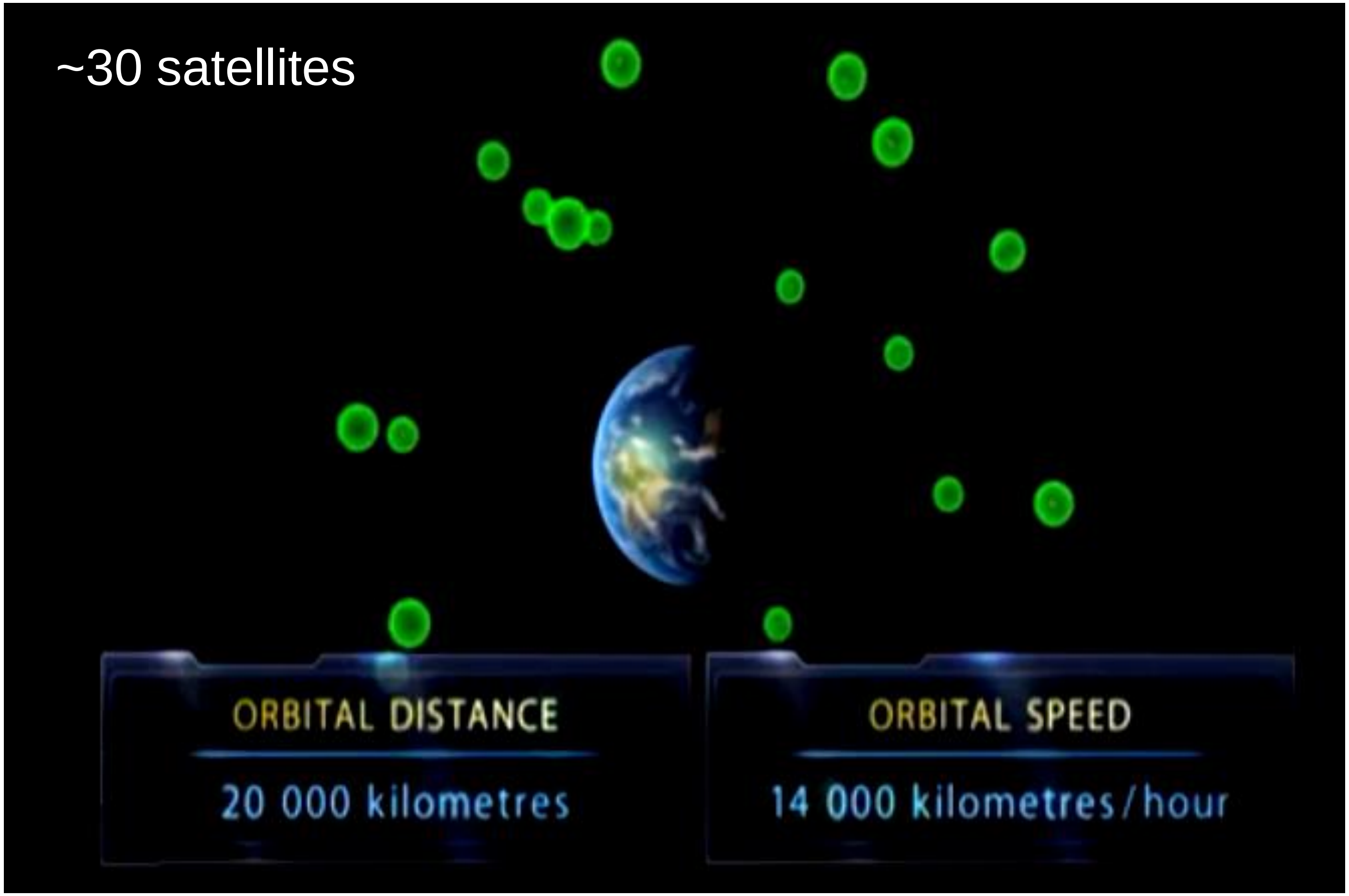
Chinese version: http://cul.sohu.com/20160220/n437974112.shtml

Laser Interferometer Gravitational Wave Observatory (LIGO)





Global positioning system (GPS) and relativity

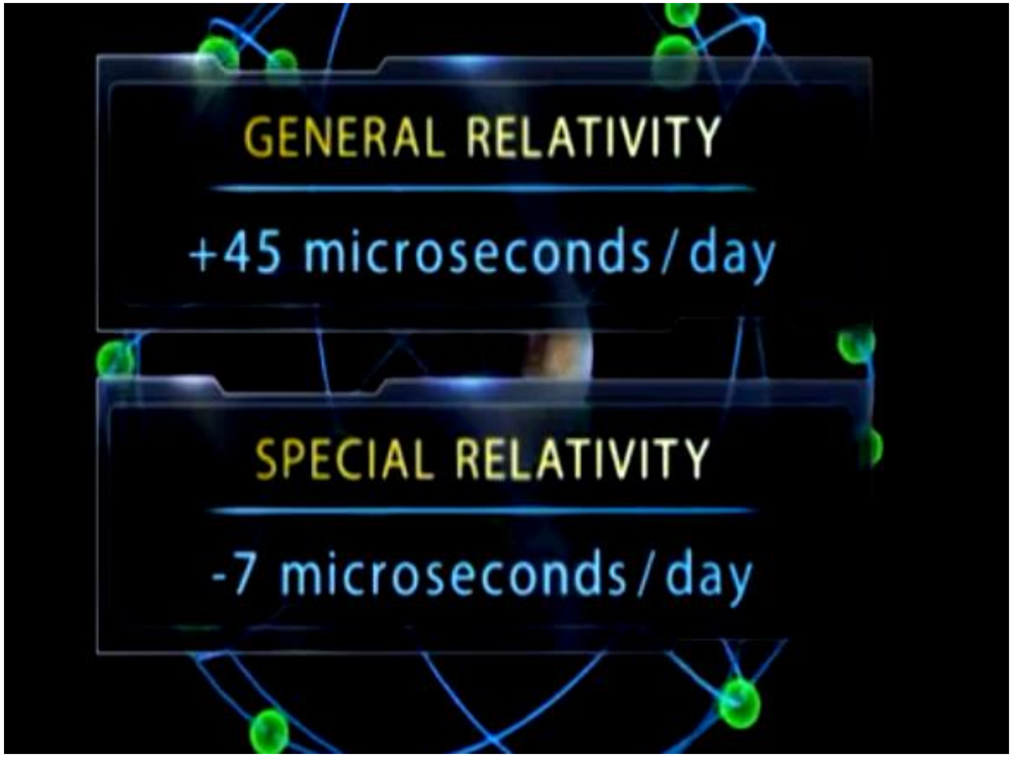


For the GPS to work, we are seeing at least 4 satellites, getting the information of where the satellite is and what time the information is sent

Global positioning system (GPS) and relativity

To accurate determine our location on earth with error smaller than a few meters, the time and distance has to calculate very precisely

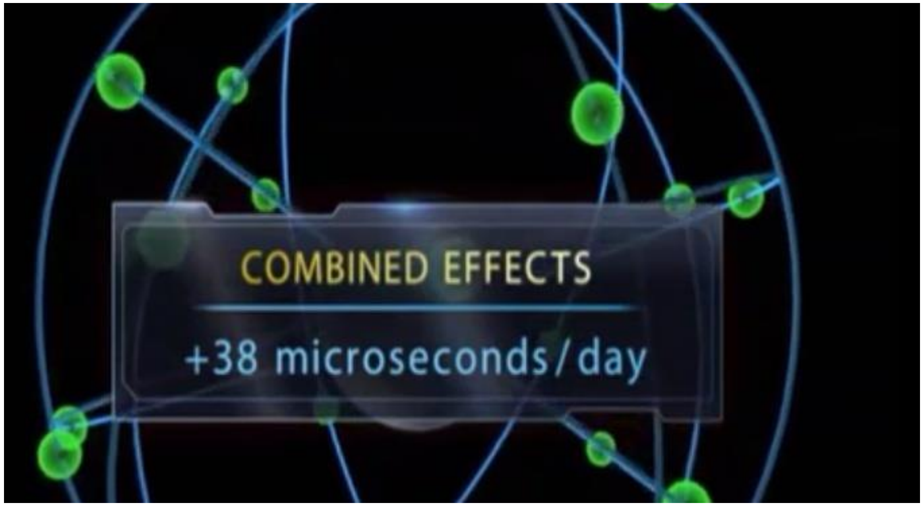
By Einstein's theory of special and general relativity, the time on the satellites are different from the time on earth.



By general relativity, clocks run faster when gravity is weaker (satellites are ~20 km away from earth surface)

By special relativity, clocks run slower as satellites are moving at a high speed (14,000 km/hr)

Global positioning system (GPS) and relativity



error of ~11 km/day
(at the speed of light)
if we did not consider
the effect of relativity !

Review & Summary

The Postulates Einstein's **special theory of relativity** is based on two postulates:

1. The laws of physics are the same for observers in all inertial reference frames. No one frame is preferred over any other.
2. The speed of light in vacuum has the same value c in all directions and in all inertial reference frames.

The speed of light c in vacuum is an ultimate speed that cannot be exceeded by any entity carrying energy or information.

Coordinates of an Event Three space coordinates and one time coordinate specify an **event.** One task of special relativity is to relate these coordinates as assigned by two observers who are in uniform motion with respect to each other.

Simultaneous Events If two observers are in relative motion, they will not, in general, agree as to whether two events are simultaneous.

Time Dilation If two successive events occur at the same place in an inertial reference frame, the time interval Δt_0 between them, measured on a single clock where they occur, is the **proper time** between the events. *Observers in frames moving relative to that frame will measure a larger value for this interval.* For an observer moving with relative speed v, the measured time interval is

$$\Delta t = \frac{\Delta t_0}{\sqrt{1-(v/c)^2}} = \frac{\Delta t_0}{\sqrt{1-\beta^2}} = \gamma\,\Delta t_0 \quad \text{(time dilation)}. \qquad \text{(37-7 to 37-9)}$$

Here $\beta = v/c$ is the **speed parameter** and $\gamma = 1/\sqrt{1-\beta^2}$ is the **Lorentz factor.** An important result of time dilation is that moving clocks run slow as measured by an observer at rest.

Length Contraction The length L_0 of an object measured by an observer in an inertial reference frame in which the object is at rest is called its **proper length.** *Observers in frames moving relative to that frame and parallel to that length will measure a shorter length.* For an observer moving with relative speed v, the measured length is

$$L = L_0\sqrt{1-\beta^2} = \frac{L_0}{\gamma} \quad \text{(length contraction)}. \qquad \text{(37-13)}$$

The Lorentz Transformation The *Lorentz transformation* equations relate the spacetime coordinates of a single event as seen by observers in two inertial frames, S and S', where S' is moving relative to S with velocity v in the positive x and x' direction. The four coordinates are related by

$$\begin{aligned} x' &= \gamma(x - vt), \\ y' &= y, \\ z' &= z, \\ t' &= \gamma(t - vx/c^2). \end{aligned} \qquad \text{(37-21)}$$

Relativity of Velocities When a particle is moving with speed u' in the positive x' direction in an inertial reference frame S' that itself is moving with speed v parallel to the x direction of a second inertial frame S, the speed u of the particle as measured in S is

$$u = \frac{u' + v}{1 + u'v/c^2} \quad \text{(relativistic velocity)}. \qquad \text{(37-29)}$$

Relativistic Doppler Effect When a light source and a light

Review & Summary

detector move directly relative to each other, the wavelength of the light as measured in the rest frame of the source is the *proper wavelength* λ_0. The detected wavelength λ is either longer (a *red shift*) or shorter (a *blue shift*) depending on whether the source–detector separation is increasing or decreasing. When the separation is increasing, the wavelengths are related by

$$\lambda = \lambda_0 \sqrt{\frac{1+\beta}{1-\beta}} \quad \text{(source and detector separating)}, \tag{37-32}$$

where $\beta = v/c$ and v is the relative radial speed (along a line connecting the source and detector). If the separation is decreasing, the signs in front of the β symbols are reversed. For speeds much less than c, the magnitude of the Doppler wavelength shift ($\Delta\lambda = \lambda - \lambda_0$) is approximately related to v by

$$v = \frac{|\Delta\lambda|}{\lambda_0} c \quad (v \ll c). \tag{37-36}$$

Transverse Doppler Effect If the relative motion of the light source is perpendicular to a line joining the source and detector, the detected frequency f is related to the proper frequency f_0 by

$$f = f_0 \sqrt{1-\beta^2}. \tag{37-37}$$

Momentum and Energy The following definitions of linear momentum $\vec{p}$, kinetic energy K, and total energy E for a particle of mass m are valid at any physically possible speed:

$$\vec{p} = \gamma m \vec{v} \quad \text{(momentum)}, \tag{37-42}$$

$$E = mc^2 + K = \gamma mc^2 \quad \text{(total energy)}, \tag{37-47, 37-48}$$

$$K = mc^2(\gamma - 1) \quad \text{(kinetic energy)}. \tag{37-52}$$

Here γ is the Lorentz factor for the particle's motion, and mc^2 is the *mass energy*, or *rest energy*, associated with the mass of the particle. These equations lead to the relationships

$$(pc)^2 = K^2 + 2Kmc^2 \tag{37-54}$$

and

$$E^2 = (pc)^2 + (mc^2)^2. \tag{37-55}$$

When a system of particles undergoes a chemical or nuclear reaction, the Q of the reaction is the negative of the change in the system's total mass energy:

$$Q = M_i c^2 - M_f c^2 = -\Delta M\, c^2, \tag{37-50}$$

where M_i is the system's total mass before the reaction and M_f is its total mass after the reaction.